



Module-3: Work & Heat

Thermodynamics (NMEC103) : (Winter 2024-25)

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Nonmechanical Work Modes of Energy Transport

Of the wide variety of nonmechanical work mode available , the following five have significant engineering value.

1. electrical current flow,
2. electrical polarization,
3. magnetic,
4. chemical, and
5. mechanochemical

- A pure conductor is a substance that has mobile charges (electron) free to move in an applied electric field. They constitute the flow of electric current.
- Pure nonconductors have no free electrons whatsoever
- semiconductor is a material that behave as dielectric (nonconductor) at low temperature but becomes conducting at high temperature

Note: As an electric field E is applied to a pure conductor the free electrons migrate to the conductors' outer surface where they create their own electric field which opposes the applied field. As more and more electrons reach the outer surface the electric field inside the object grows weaker and weaker, eventually vanishing altogether. At equilibrium, there is no electric field within a pure conductor

- A pure **nonconductor** has no free electrons with which to neutralize the applied electric field.
- The externally applied field therefore acts on the internal molecules and normally **nonpolar molecules will become polar and develop electric dipoles**. Some **molecules are naturally polar in the absence of electric field (e.g. , water)**.
- The applied electric field will **rotate and align the newly created or naturally polar molecules**. Complete alignment is normally prevented by molecular vibration. But when the applied field is strong enough to overcome the vibration randomizing effects and further increase in the field strength has no effect on the material , the material is then said to be saturated with the applied field.
- The process of **electric dipole creation**, **rotation**, and **alignment in an applied electric field** is known as **dielectric polarization**.

Therefore, there are **two separate work modes that arise from the application of an electric field to the material**. The first is the work associated with the **free electrons (current) flow**, and the **second is the work associated with the dielectric polarization**. For **pure conductor polarization work is zero**, and for a **pure nonconductor the current flow work is always zero**

Electrical Current Flow Work

Electrical current flow work occur when a current carrying wire crosses the system boundary. The generalized force here is the intensive property voltage (the electric potential) ϕ and the extensive property generalized displacement is the charge q . Then

$$(\delta W)_{\text{electrical current}} = \phi dq$$

$$(W_{1-2})_{\text{electrical current}} = \int_1^2 \phi dq$$

Electric current i is defined as

$$i = \frac{dq}{dt}$$

$$(\delta W)_{\text{electrical current}} = \phi i dt$$

$$(W_{1-2})_{\text{electrical current}} = \int_1^2 \phi i dt$$

From the Ohm's law the instantaneous voltage ϕ across a pure resistor R carrying an alternating current described by $i = i_{\max} \sin(2\pi ft)$ is

$$\phi = Ri = R i_{\max} \sin(2\pi ft)$$

Where f is the frequency and $\phi_{max} = Ri_{max}$

The electrical current work of n cycles of an alternating electrical current passing through a pure resistance from time 0 to $t = n/f$ as

$$(W_{1-2})_{electrical\ current} = \phi_{max} i_{max} \int_0^{n/f} \sin^2(2\pi ft) dt = \phi_{max} i_{max} (t/2)$$

$$(W_{1-2})_{electrical\ current} = \phi_e i_e t = \phi_e^2 (t/R) = i_e^2 R t$$

ϕ_e and i_e are the effective voltage and current defined as

$$\phi_e = \phi_{max} / \sqrt{2} \quad i_e = i_{max} / \sqrt{2}$$

The electrical potential ϕ and the electric field strength vector $\mathbf{E}$ and related by

$$\mathbf{E} = -\nabla(\phi)$$

Electric work can exist in either open or closed system (we do not consider the flow of electrons across a system boundary to be a mass flow term). When the electron supply is coming from or going into a finite energy reservoir such as battery or capacitor then the following integral equation is used

$$(W_{1-2})_{\text{electrical current}} = \int_1^2 \phi dq = \int_1^2 \phi i dt$$

When an essentially infinite reservoir with constant voltage and current is used then it is more convenient to use instantaneous electrical power, defined as

$$(\dot{W})_{\text{electrical current}} = \frac{dW}{dt} = \phi i$$

The instantaneous electrical power ϕi of an alternating current circuit will vary in time with the excitation frequency f . However, it is common to report the electrical power of an ac device as the instantaneous power averaged over one cycle of oscillation, or

$$(\dot{W})_{\text{electrical current}} = f \int_0^{1/f} \phi i dt = f \phi_{\max} i_{\max} \int_0^{1/f} \sin^2(2\pi f t) dt = i_e^2 R$$

Consider a 120 volt and 144 ohm alternating current incandescent light bulb is to be a pure resistance. Determine

- a) The electric current work when the bulb is operated for 1.5 h and
- b) The electric power consumption

Since the voltage and current rating of an ac device are always given in terms of their effective values, then $\phi_e = 120\text{ V}$ and from Ohm's law $i_e \frac{\phi_e}{R} = \frac{120}{144} = 0.833\text{ A}$

$$(W_{1-2})_{\text{electrical current}} = \phi_e i_e t = (120\text{ V})(0.833\text{ A})(1.5\text{ h}) = 150\text{ W.h}$$

$$(\dot{W})_{\text{electrical current}} = \phi_e i_e = (120\text{ V})(0.833\text{ A}) = 100\text{ W}$$

Electrical Polarization Work

The electric dipole formation , rotation, and alignment that occur when an electric field is applied to a nonconductor or a semiconductor constitutes an electrical polarization work mode. The generalized force is the intensive property E , the electric field strength vector , and the generalized displacement is the extensive property P , the polarization vector of the medium (defined to be the sum of electric dipole rotation moments of all molecules in the system)

$$(\delta W)_{\text{electrical polarization}} = \mathbf{E} \cdot d\mathbf{P}$$

$$(W)_{1-2, \text{electrical polarization}} = \int_1^2 \mathbf{E} \cdot d\mathbf{P}$$

Since the effect of the electric field is to orient the dipoles coincident with the field, then E and P are always parallel and point in the same direction

$$(W)_{1-2, \text{electrical polarization}} = \int_1^2 E dP$$

Many substances (particularly gases) correlate well with the following dielectric equation of state:

$$P = \forall \epsilon_0 \chi_e E$$

$\forall$ = Volume of the dielectric substance χ_e is the electric susceptibility (non-dimensional) of a material

ϵ_0 is the electric permittivity in vacuum $\left(8.85419 \times \frac{10^{-12} \text{ N}}{\text{V}^2} \right)$

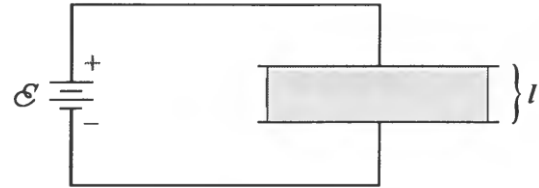
A parallel plate capacitor is charged to a potential difference of 110 V at 25°C. The plates are square with a side lengths of 0.10m and are separated by 0.01 m. If the gap between plates are filled with water, determine the polarization work required to charge the capacitor.

$$(W)_{1-2, \text{electrical polarization}} = \int_1^2 \mathbf{E} \cdot d\mathbf{P}$$

$$(W)_{1-2, \text{electrical polarization}} = \int_1^2 E dP = \int_1^2 V \epsilon_0 \chi_e E dE$$

$$(W)_{1-2, \text{electrical polarization}} = V \epsilon_0 \chi_e (E_2^2 - E_1^2) / 2$$

$$V = AL = (0.10 \text{ m})^2 (0.010 \text{ m}) = 10^{-4} \text{ m}^3$$



If we assume that electric potential ϕ varies linearly between the plates, then we can

$$E = |-\nabla(\phi)| = (\text{Voltage difference}) / (\text{plate gap})$$

$$E_1 = 0 \text{ (uncharged plates) and}$$

$$\text{So that } E_2 = \frac{110}{0.01} = 1.1 \times \frac{10^4 \text{ V}}{\text{m}}$$

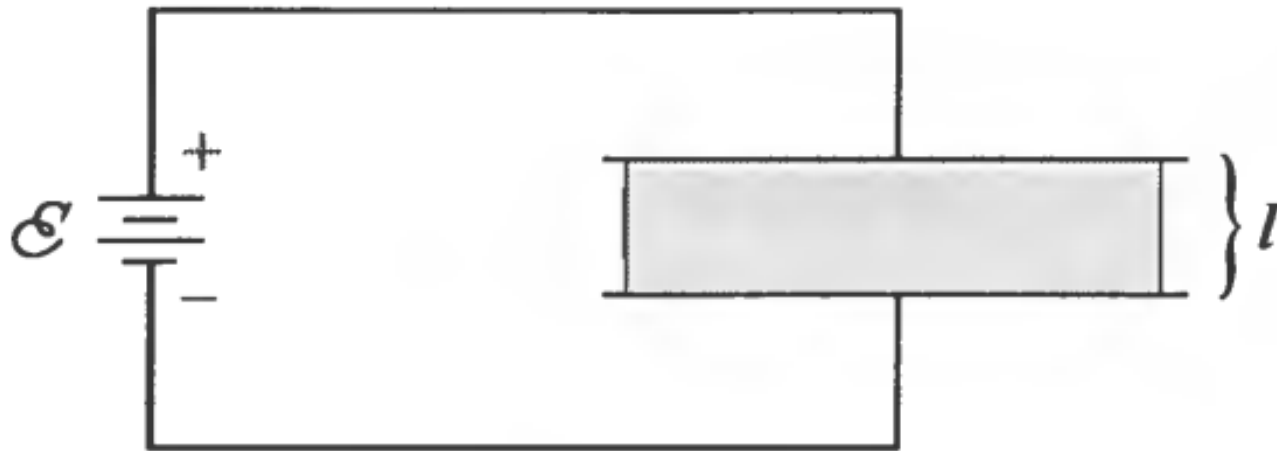
We find that for water, $\chi_e = 77.5$

$$(W)_{1-2, \text{electrical polarization}} =$$

$$(10^{-4} \text{ m}^3) (8.85419 \times 10^{-12} \text{ N/V}^2) (77.5) \times \left[(1.1 \times 10^4)^2 - 0 \right]$$

Note:

- ❑ The polarization work is a small fraction of the total energy required to charge an entire capacitor.
- ❑ The total work required to charge a capacitor comprises of two parts.
- ❑ The largest fraction goes into increasing the electric field strength itself. And the remaining goes into the polarization of material exposed to the electric field
- ❑ If the thermodynamic system we are analysing is just the material between the plates of a capacitor, then only the polarization work is done on the material given by $\int_1^2 \mathbf{E} \cdot d\mathbf{P}$ (gives the correct electrical work mode)
- ❑ If we analyse the entire capacitor (plates and dielectric) then the following equation must be used to estimate correct electrical work mode value. $\int_1^2 \phi i dt$



Magnetic Work

Materials are classified as:

- **Diamagnetic** materials have no permanently established molecular magnetic dipoles. However, when they are placed in a magnetic field, their molecules develop magnetic dipoles whose magnetic field opposes the applied field (the Greek prefix *dia* means “to oppose”).
- **Paramagnetic** materials have naturally occurring molecular magnetic dipoles. When placed in a magnetic field, these dipoles tend to align themselves parallel to the field (the Greek prefix *para* means “beside”).
- **Ferromagnetic** materials retain some magnetism after the removal of a magnetic field. The thermodynamic state of these materials depends not only on the present values of their thermomagnetic properties, but also on their magnetic history. In this sense, ferromagnetic materials have a “memory” of their previous magnetic exposure.

As in the case of an electric field, the work associated with the initiation or destruction of a magnetic field consists of two parts. The first part is the work required to change the magnetic field itself (as though it existed within a vacuum), and the second part is the work required to change the magnetization of the material present inside the magnetic field.

For the calculation of the work the **generalized force is the intensive property** called the **magnetic field strength $\vec{H}$ (A/m²)** and the **generalized displacement is the extensive property $V\vec{B}$** , the product of the **system volume V (m³)** and the **magnetic induction $\vec{B}$** (in tesla or V.s/m²). Thus, assuming the magnetic field is applied to the system,

$$\delta W_{magnetic} = -\vec{H} \cdot d(V\vec{B})$$

$\vec{H}$ & $\vec{B}$ are always parallel and point in the same direction in magnetic materials, this reduces to

$$\delta W_{magnetic} = -Hd(VB)$$

The magnetic induction can be decomposed into two vectors as:

$$\vec{B} = \mu_o \vec{H} + \mu_o \vec{M}$$

Where $\vec{M}$ is the magnetisation vector per unit volume of the material exposed to the magnetic field. (In vacuum, $\vec{M}$ is a null vector $\vec{0}$) $\mu_o = 4\pi \times 10^{-7} \text{ V.s/(Am)}$ is a universal constant called the *magnetic permeability*

$$\delta W_{magnetic} = -H \cdot d(V\mu_o \vec{H} + V\mu_o \vec{M})$$

$$\delta W_{magnetic} = -\mu_o H \cdot d(V\vec{H}) - \mu_o H \cdot d(V\vec{M})$$

The first term corresponds to the work required to just change the field itself (in a vacuum) and the 2nd term is the work associated with the alignment of magnetic dipoles of the material present inside the magnetic field, and represents the work of magnetisation of the material exposed to the magnetic field. Thus, we can write

$$\delta W_{material\ magnetization} = \mu_o H \cdot d(V\vec{M})$$

A simple and useful equation of state for a magnetic field is $M = \chi_m H$

χ_m is the magnetic susceptibility (a dimensionless number)

$$\delta W_{\text{magnetic}} = -\mu_o H \cdot d(VH) - \mu_o H \cdot d(V\chi_m H)$$

$$\delta W_{\text{magnetic}} = -\mu_o V(1 + \chi_m) \cdot H dH$$

For constant volume and constant magnetic susceptibility

$$W_{1-2,\text{magnetic}} = -\mu_o V(1 + \chi_m) \left(\frac{H_2^2 - H_1^2}{2} \right)$$

$$W_{1-2,\text{material magnetisation}} = -\mu_o V \chi_m \left(\frac{H_2^2 - H_1^2}{2} \right)$$

Chemical Work:

Generalised Force = μ_i , **Gibb's potential of the i^{th} species**

Generalised Displacement: m_i , mass of the chemical species added or removed

Work done during mixing/separation of chemical species:

$$(W)_{1-2} = \int_1^2 \sum_{i=1}^k \mu_i dm_i$$

When the **chemical potential** is constant during the mass transfer from state 1 to state 2.

$$(W)_{1-2} = \sum_{i=1}^k \mu_i (m_2 - m_1)_i$$

Important Note: Chemical work **does not include the energy transports produced by chemical reaction**, nor does it include the energy transported across the system boundary with the mass transport itself. **The chemical work presented here essentially deals only with those energy transports involved in the mixing or separating**

Mechanochemical Work: Mechanochemical work occurs whenever there is a direct conversion from chemical to mechanical energy. Animal muscles are the best example of that. The fuel used in mechanochemical work is not burned, as in standard heat engine. Often it is merely diluted and a small amount of chemical work is simultaneously extracted.

$$\delta W = f dl$$

$$(W)_{1-2} = \int_1^2 f dl$$

f is the generalised force (intensive) and l is the generalised displacement (extensive) for mechanochemical system

Simple System	Intensive Coordinate (Generalized Force)	Extensive Coordinate (Generalized Displacement)	Work
Simple compressible system	P (pa)	V, in m ³	$(W)_{1-2} = \int_1^2 P dV$
Stretched Wire	T (N)	L, m	$(W)_{1-2} = - \int_1^2 T dL$
Surface Film	γ (N/m)	A, m ²	$(W)_{1-2} = - \int_1^2 \gamma dA$
System in which work is completely electrical (electrochemical cell)	E (Volt)	Z (C) Coulomb	$(W)_{1-2} = - \int_1^2 E dZ$
Paramagnetic solid	$\mu_o H$ (N/A-m)	$\mathcal{M}$, in Am ²	$(W)_{1-2} = \mu_o \int_1^2 H \cdot d\mathcal{M}$

The general form for work in reversible process for closed system

$$\delta W = (\text{intensive property}) d(\text{extensive property}) \quad W_{1-2} = \int_1^2 F_{gen} x_{gen}$$

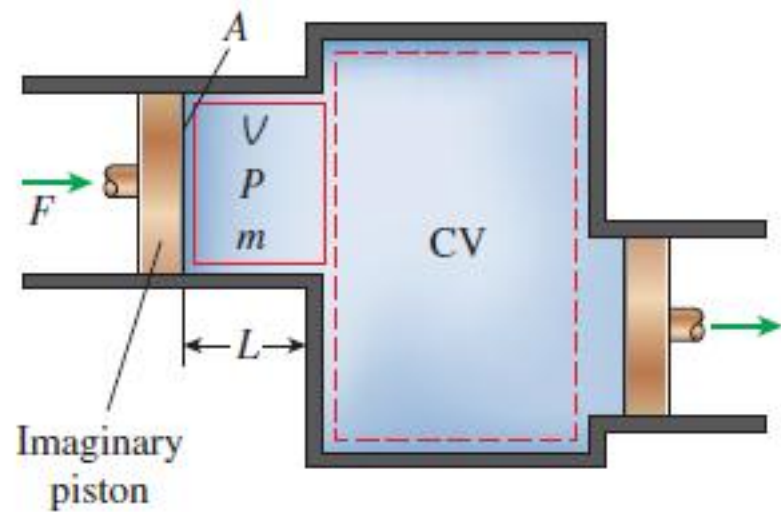
Work mode	Power Equation
Mechanical moving boundary	$\dot{w}_{moving\ boundary} = p \frac{dv}{dt} = p \dot{v}$
Mechanical Rotating Shaft	$\dot{w}_{rotating\ shaft} = T \cdot \frac{d\theta}{dt} = -T \cdot \omega$
Mechanical elastic	$\dot{w}_{elastic} = \sigma V \frac{d\epsilon}{dt} = -\sigma V \dot{\epsilon}$
Mechanical Surface tension	$\dot{w}_{surface\ tension} = -\gamma \frac{dA}{dt}$
Electrical current	$\dot{w}_{electrical\ current} = \phi i$
Electric Polarization	$\dot{w}_{electrical\ polarization} = E \left(\frac{dP}{dt} \right) = -E \dot{P}$
Magnetic	$\dot{w}_{magnetic} = -\mu_o V (1 + \chi_m) H \left(\frac{dH}{dt} \right)$ $= \mu_o V (1 + \chi_m) H \dot{H}$
Chemical	$\dot{w}_{chemical} = - \sum \mu_i \left(\frac{dm_i}{dt} \right) = \sum \mu_i \dot{m}_i$
Mechanochemical	$\dot{w}_{Mechano\ chemical} = -f \frac{dl}{dt} = f \dot{l}$

A system may be exposed to only one of the discussed work modes of energy transport or it may be exposed several mode. Since work is an additive quantity, to get the total (net work) of a system that has more than one work mode present we simply add all these work terms together.

$$\begin{aligned} \delta W &= PdV - Td\theta - \sigma d\epsilon - \gamma dA - \phi idt - EdP + \mu_0 Hd(VM) \\ &+ \sum_{i=1}^k \mu_i dm_i + fdl \end{aligned}$$

Estimation of flow work:

Whenever a fluid enters a control volume at state i or exits at state e , there is a boundary movement work associated although there may not be any real boundary present there. Consider an amount of mass at the inlet as shown in the figure flowing into the control volume. As this mass flows in there is a pressure at its back surface, so as this mass moves into the control volume it is being pushed by the mass behind it (thus fluid upstream may be imagined to be a piston), which is the surrounding. The net effect is that after the mass has entered the control volume the surrounding has pushed it against the local pressure with a velocity giving it a rate of work



$$\dot{W}_{flow} = \frac{FL}{t} = FVel = \int pVel dA = p\dot{V} = pv\dot{m}$$

$$\dot{W}_{flow,i} = p_i(Vel_i A_i) = p_i \dot{V}_i = (-) p_i (v_i \dot{m}_i)$$

$$\dot{W}_{flow,e} = p_e(Vel_e A_e) = p_e \dot{V}_e = + p_e (v_e \dot{m}_e)$$

$$\text{specific flow work} = pv$$

Vel – Velocity

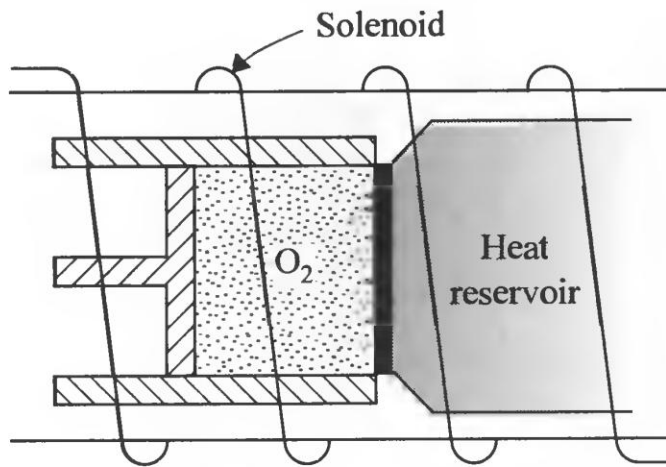
$\dot{V}$ – Volume flow rate

v – Specific volume

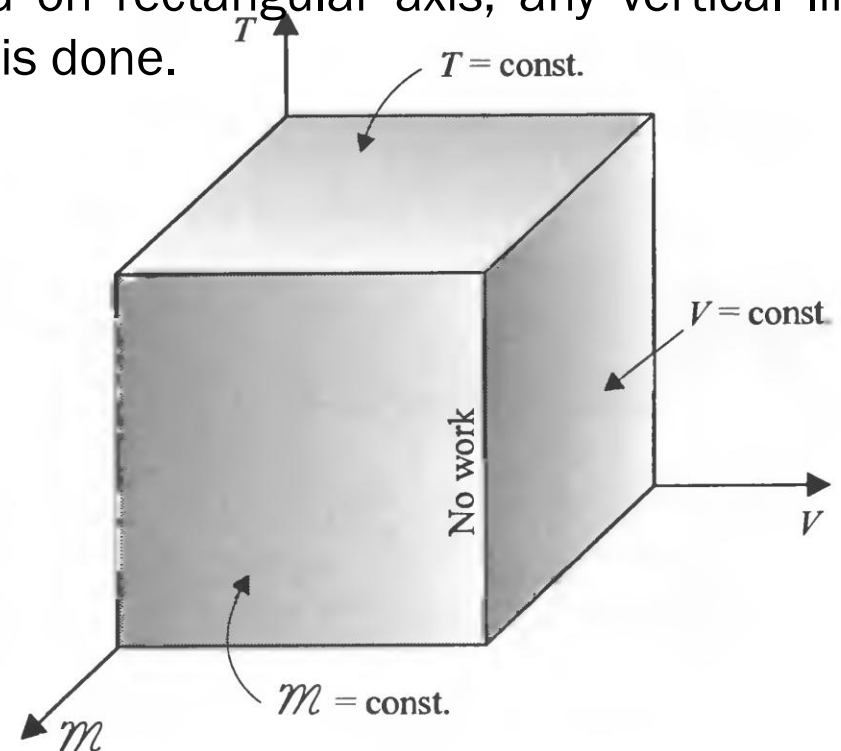
Here p is the local upstream pressure, Vel is the local velocity, and $\dot{V}$ is the volume flowrate. The flow work is also called as pressure energy, unlike the internal and kinetic energies a fluid can not store pressure energy. Fluid is a carrier of the pressure energy, like a belt transmits energy from the driving pulley to driven pulley,

Composite System (Not a Simple System)

Consider an ideal **paramagnetic gas such as oxygen** at low pressure as depicted in the figure. The oxygen may have its pressure P and volume V varied by a piston cylinder combination, and it is **submerged in a magnetic field** which may be changed by varying the current in the surrounding solenoid. The coordinates are $p, V, \mu\mathcal{H}, m, T$, only three of which are independent because of two equation of state: ideal gas equation of state $pV = nRT$, Curie's law $m = C_c \mathcal{H}/T$. Since the work done in any infinite process is $\delta W = pdV - \mu_o \mathcal{H} dm$, the most convenient coordinates are T, V, m . Which are plotted on rectangular axis, any vertical line would represent a process in which no work is done.



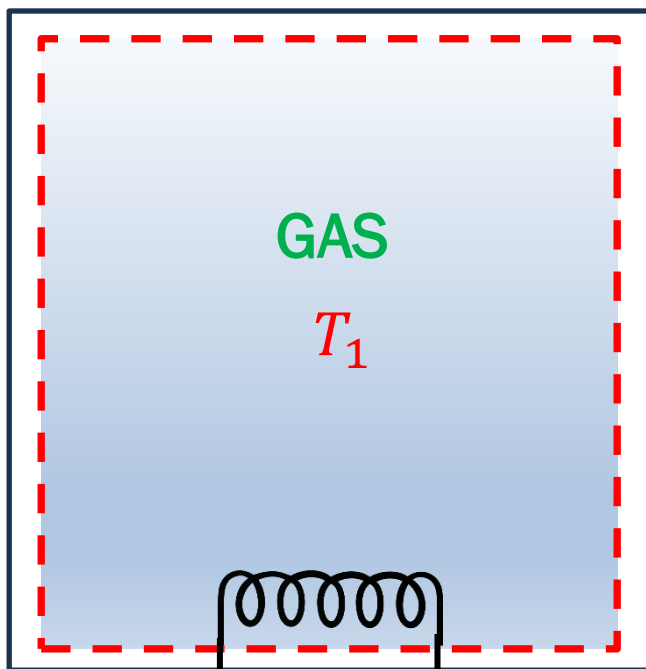
(a)



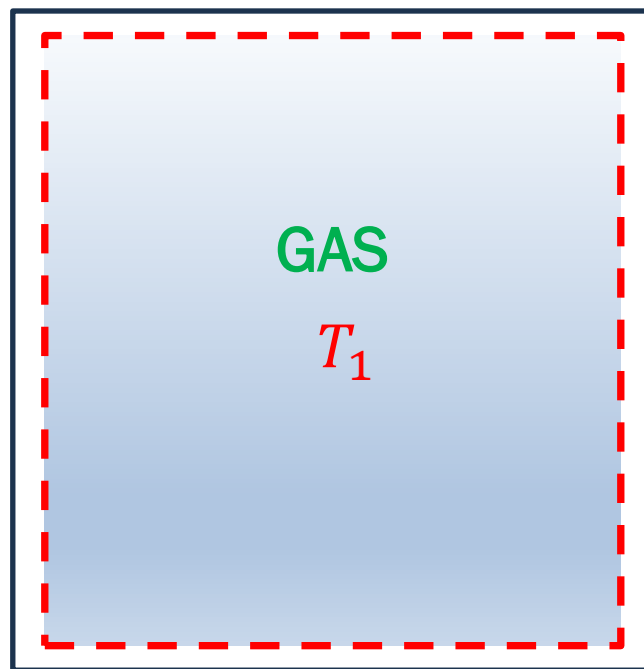
(b)

Heat: Mode of Energy Transfer

What is Heat? How to distinguish between heat and work? The simplest way to define heat transport energy is as any energy transport that is neither a work mode nor a mass flow energy transport mode. Although many authors define heat as an energy in transit due to temperature difference, this definition is not fully correct. For example the flow of electric current in certain electrical circuits can cause large heating or cooling effects. In this case the heat transfer is brought about by the electrical current flow rather than by temperature difference, and the coupling effect is called thermoelectricity. This effect can also be reversed and heat transport of energy can be converted directly (at less than 100% efficiency) into electricity. The heat transfer is broken into two fundamental modes i.e., conduction and radiation. Convection is really mass flow energy transport mode. Convection is advection (bulk fluid motion) assisted conduction and/or radiation. It is actually not the heat but the internal energy and bulk kinetic energy (i.e., stored energy of a fluid) get convected.

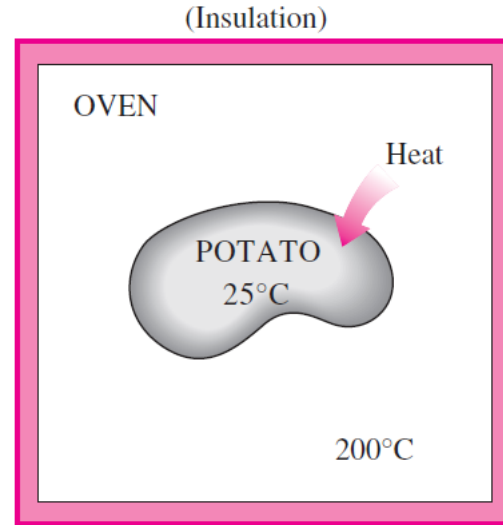


Electric work

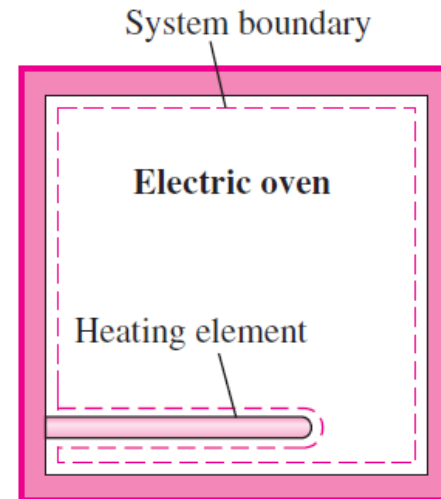
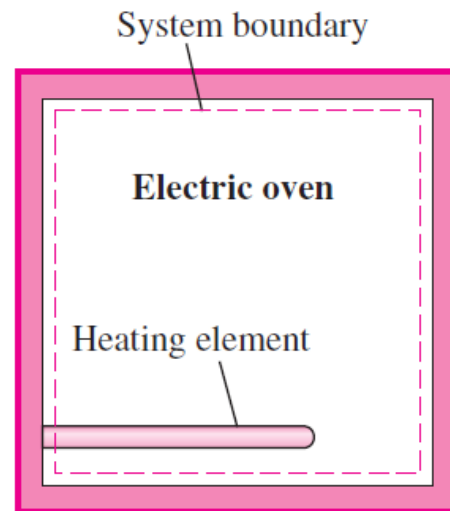


Heat Interaction

A potato initially at room temperature (25°C) is being baked in an oven that is maintained at 200°C , as shown in Fig. Is there any heat transfer during this baking process?

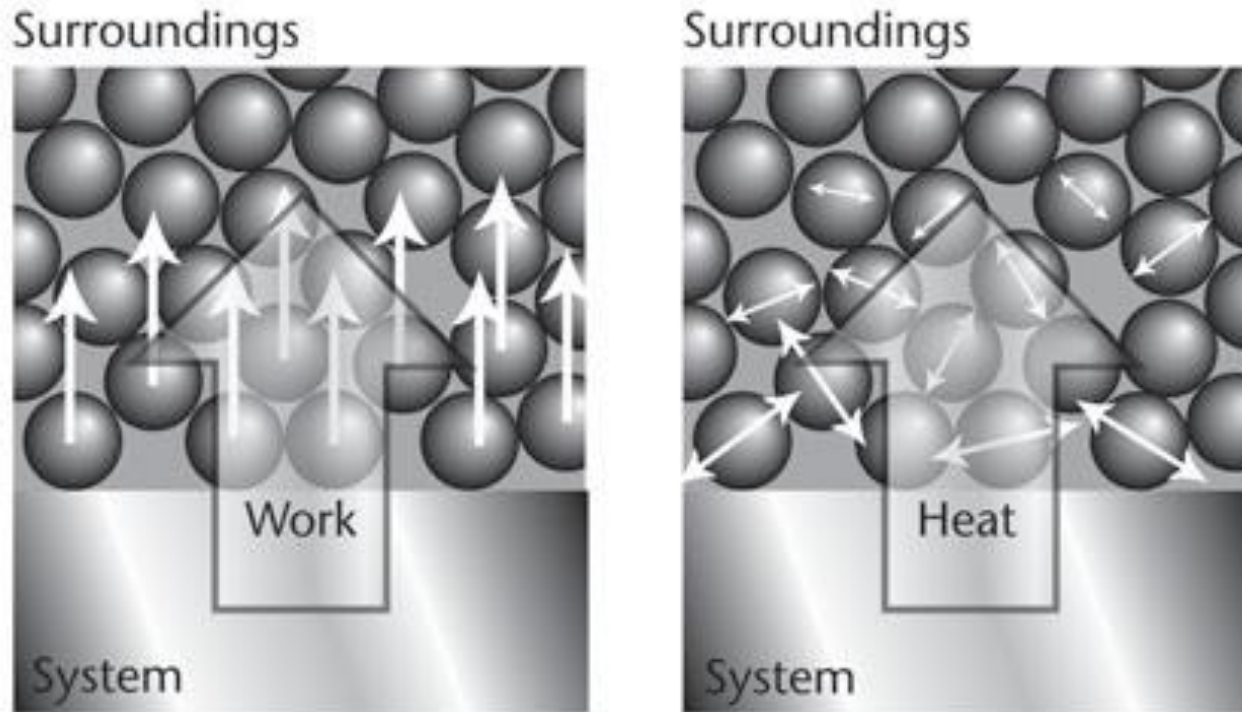


A well-insulated electric oven is being heated by its heating element. It is to be determined whether this is a heat or work interaction. Answer the question if the system is taken as only the air in the oven without the heating element.



What is Heat? How to distinguish between heat and work?

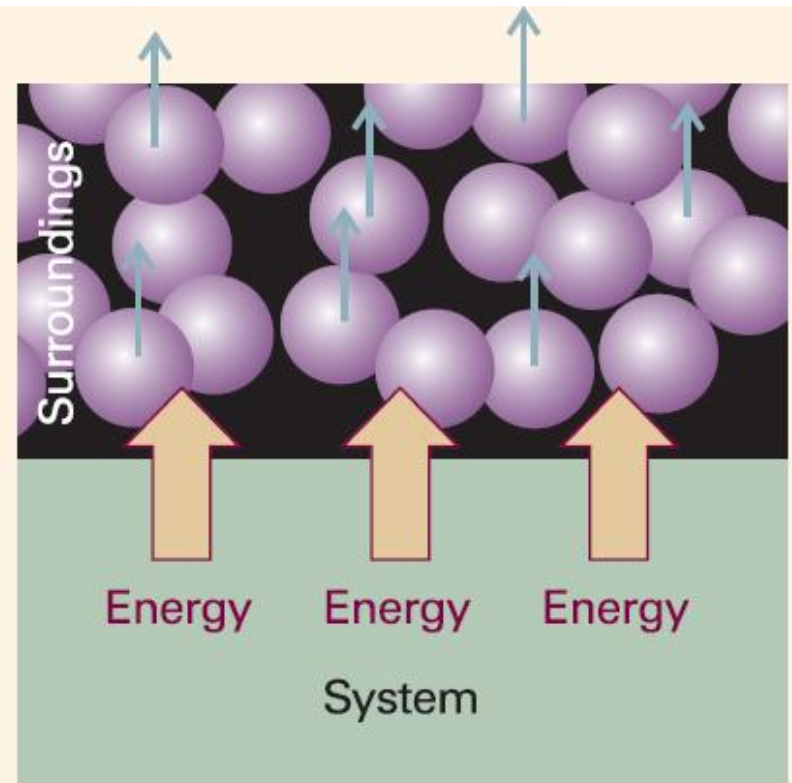
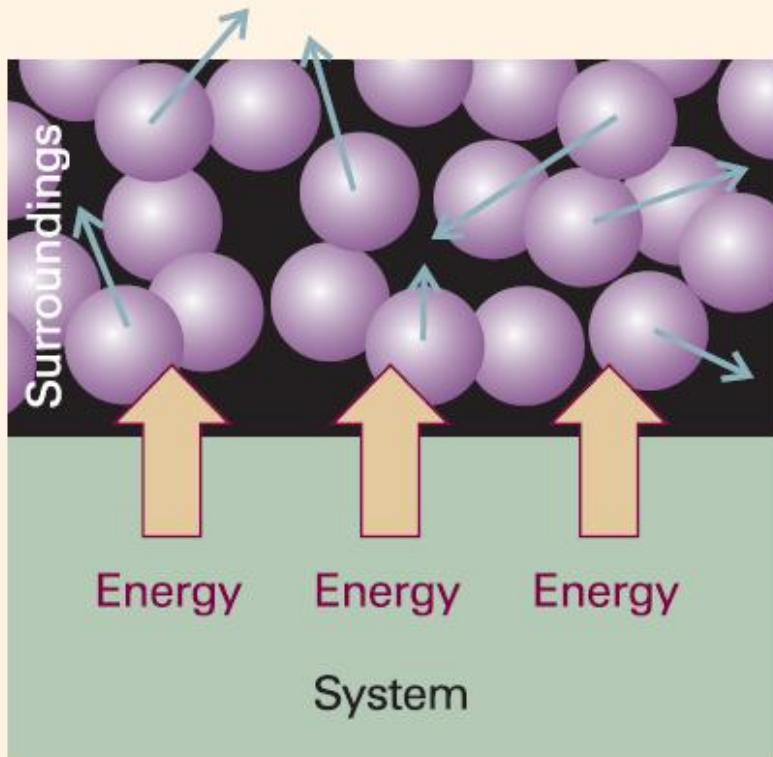
Heat is a energy in transit due to temperature difference. It is a low grade energy (disordered form of energy). All other energy in transit which transfers not because of temperature difference can be taken as work.



$Q > 0$: heat transfer *to* the system

$Q < 0$: heat transfer *from* the system

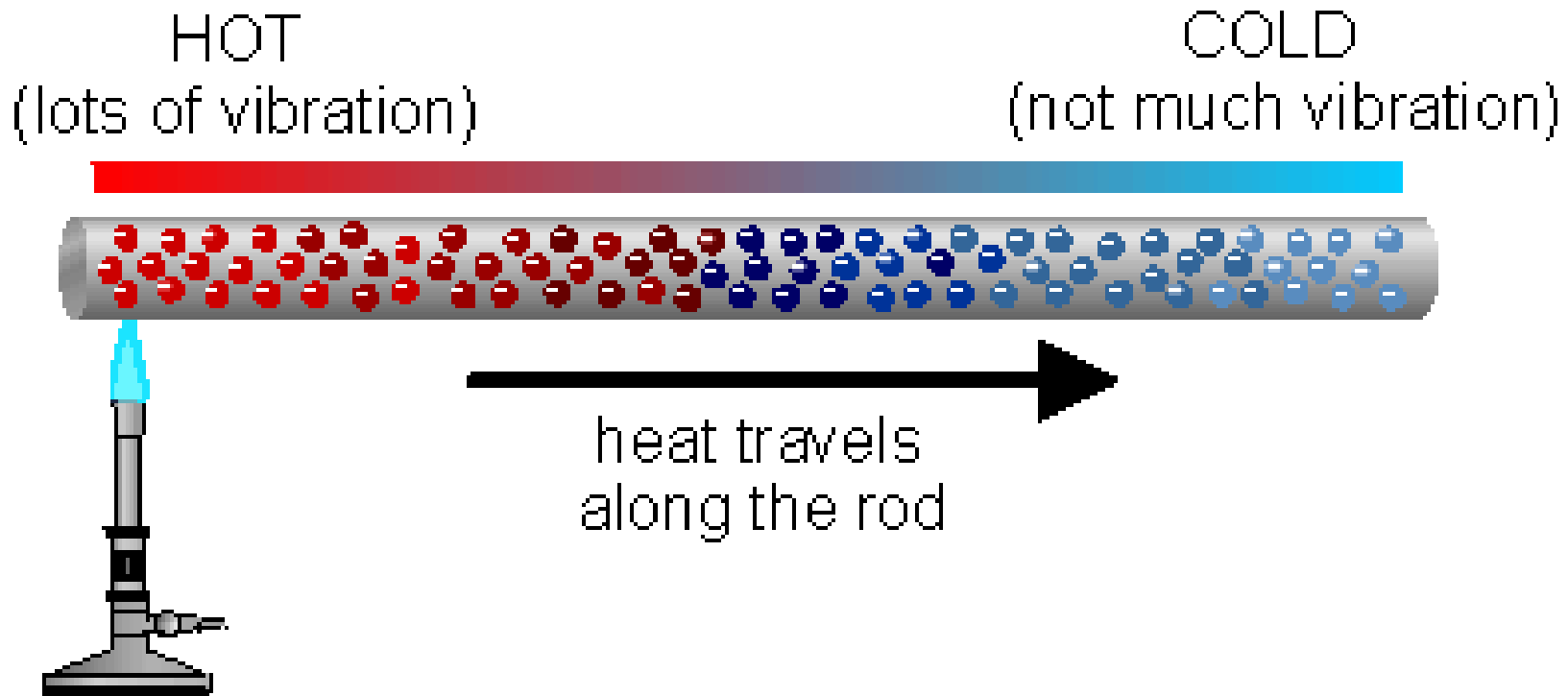
Sign Convention



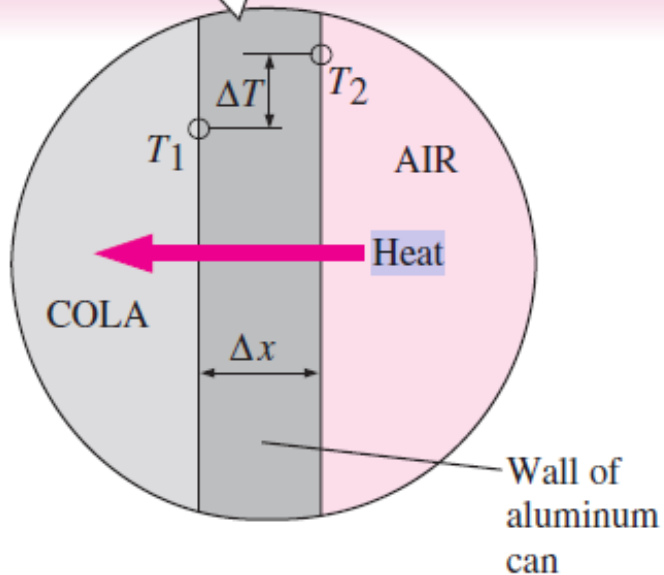
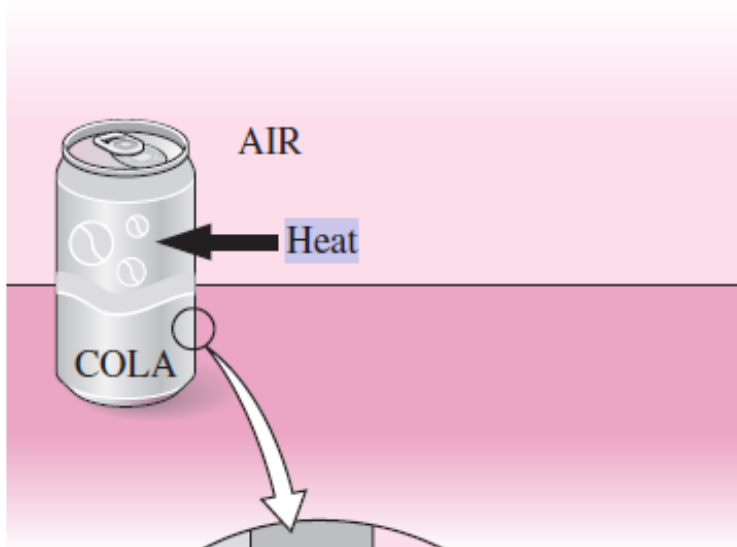
Modes of Heat Transfer : Thermodynamic is silent about the mechanism of heat transfer which includes the dynamics

Thermal Conduction

Fourier's Constitutive Relation $\dot{Q} = -kA \frac{dT}{dx}$



At microscopic level both work and heat are work mode of energy transfer

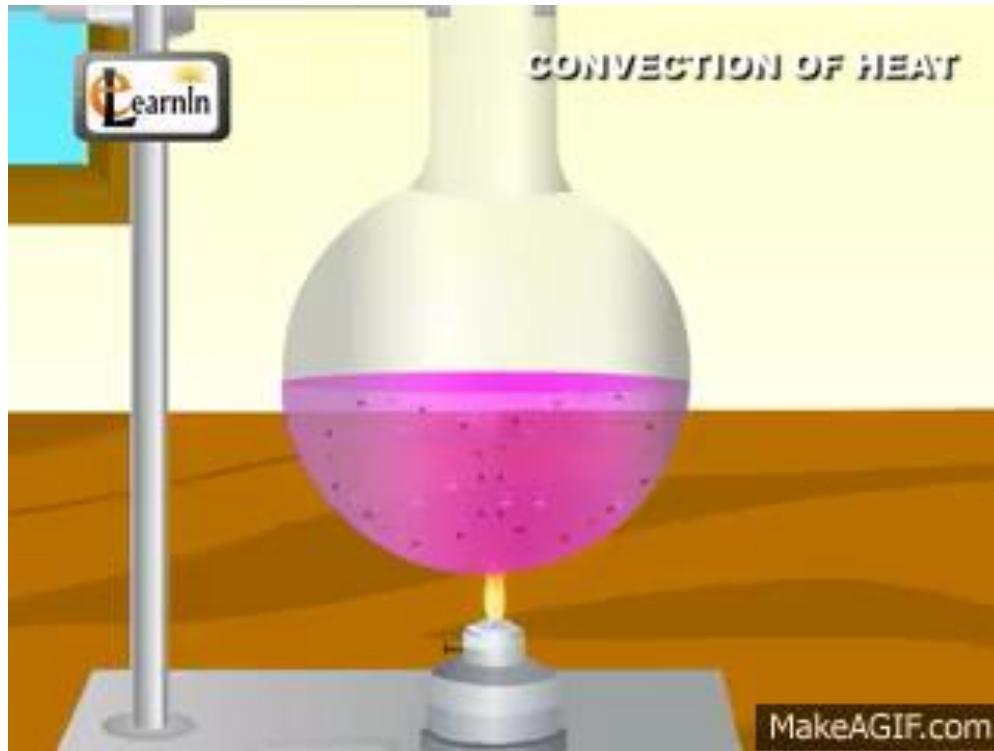


$$\dot{Q} = -kA \frac{dT}{dx} = -kA \frac{T_2 - T_1}{\Delta x}$$

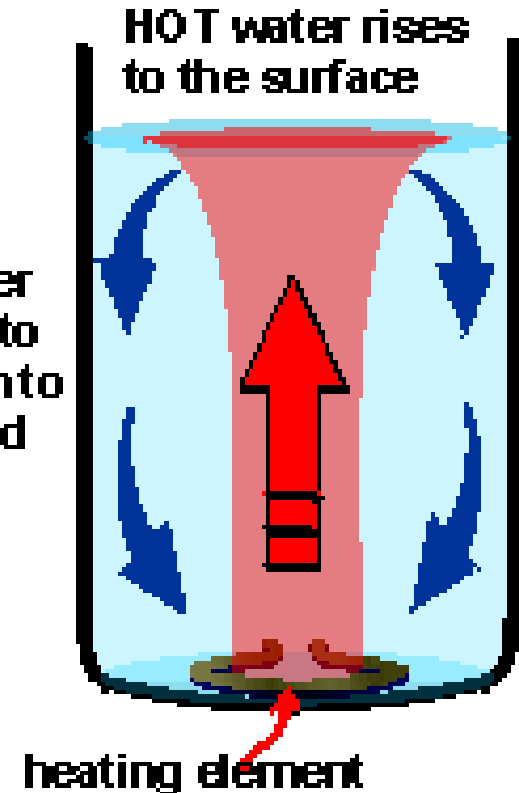
Thermal conductivities of some materials at room conditions

Material	Thermal conductivity, W/m · K
Diamond	2300
Silver	429
Copper	401
Gold	317
Aluminium	237
Iron	80.2
Mercury (ℓ)	8.54
Glass	1.4
Brick	0.72
Water (ℓ)	0.613
Human skin	0.37
Wood (oak)	0.17
Helium (g)	0.152
Soft rubber	0.13
Glass fiber	0.043
Air (g)	0.026
Urethane, rigid foam	0.026

Convection



COLD water
circulates to
the bottom to
be reheated



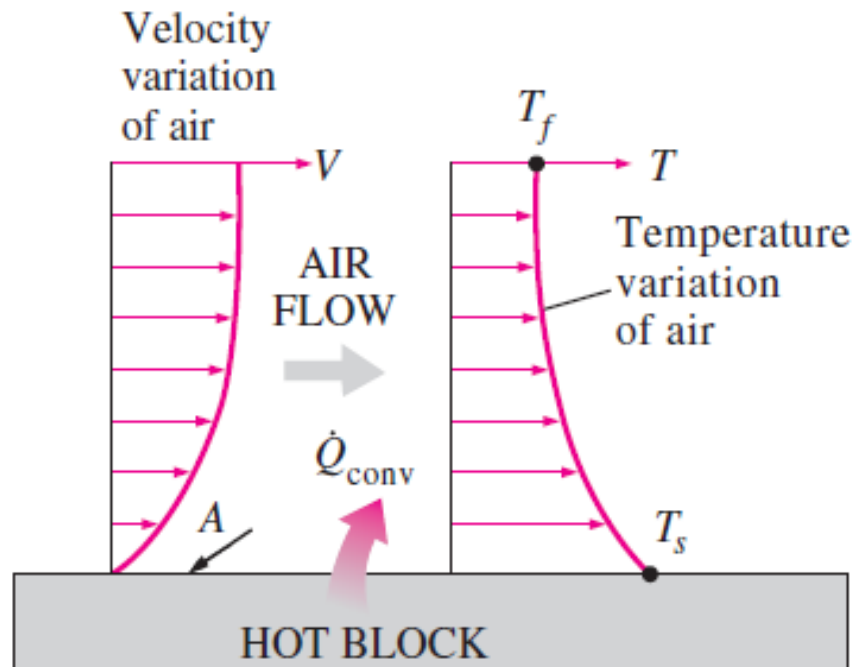
Newtons Law of Cooling: $\dot{Q}_{conv} = hA(T_s - T_{\infty})$

Convection is called **forced** if the fluid is forced over a surface or through a tube by an external means such as a **fan, blower, pump or the wind**

In contrast convection is called **free or natural** if the fluid motion is caused by the **buoyancy force** induced by the **density gradient caused by temperature gradient or concentration gradient**.

Heat transfer process that involve the **change of phase** of fluid also considered to be convection because of the **fluid motion induced during the process such as the rise of the vapor bubble during boiling** and **fall of the liquid droplets during condensation**





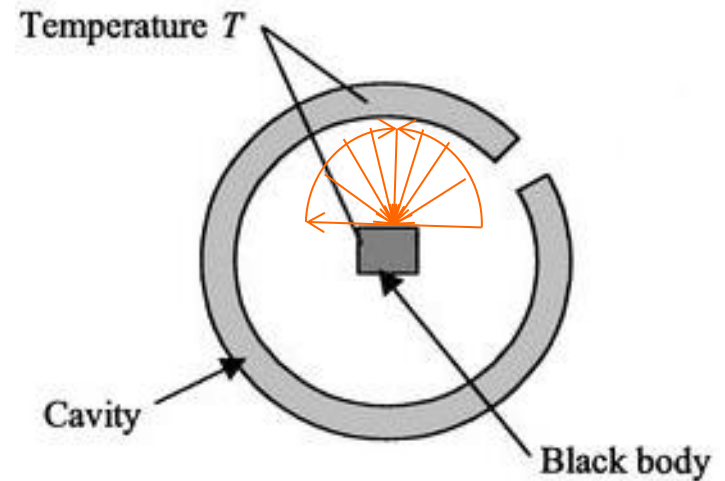
$$\dot{Q}_{conv} = hA(T_s - T_f)$$

The h value depends on surface geometry, the nature of fluid motion, the properties of the fluid, and the bulk fluid velocity.

Typical values of h , in $\text{W/m}^2\text{-K}$, are in the range of 2–25 for the free convection of gases, 50–1000 for the free convection of liquids, 25–250 for the forced convection of gases, 50–20,000 for the forced convection of liquids, and 2500–100,000 for convection in boiling and condensation processes

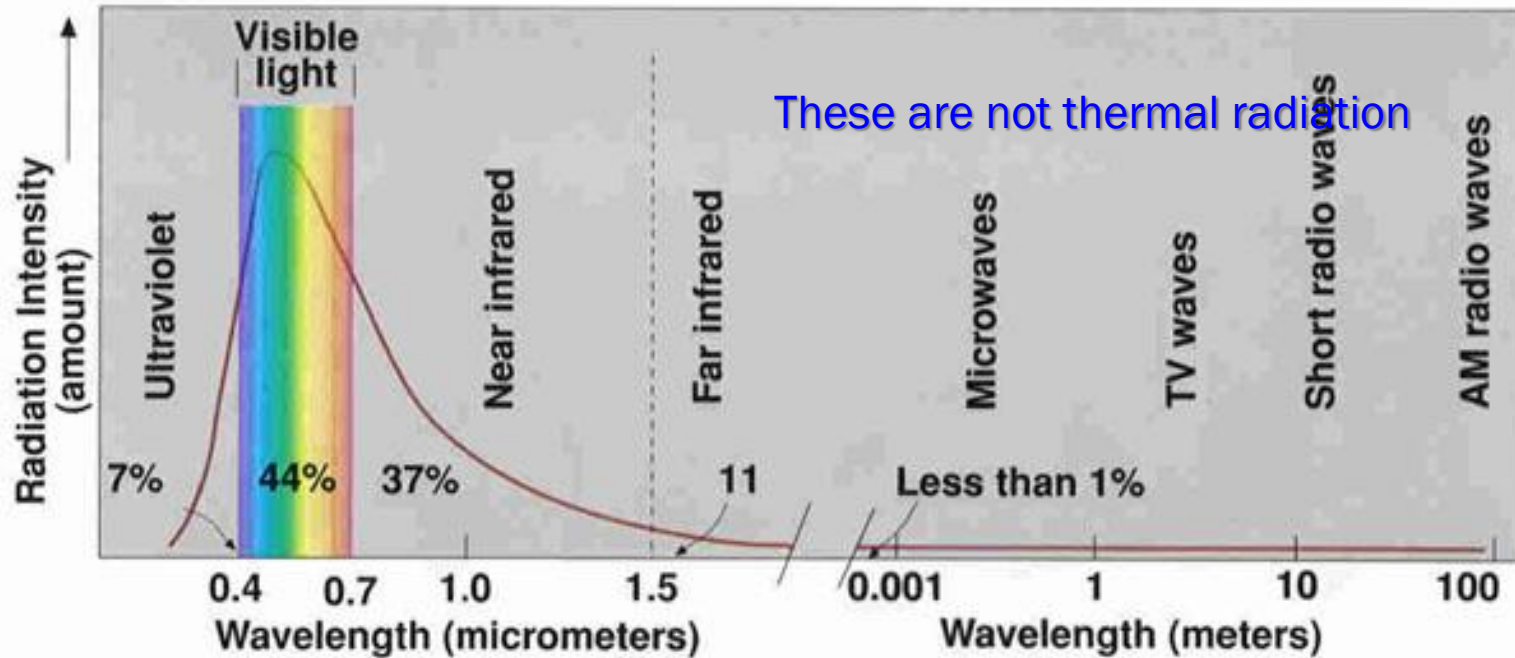
Radiation

- Heat transfer by **electromagnetic waves (photons)** (moves at the **speed of light** as light is an electromagnetic wave), The heat transfer by radiation is function of **wavelength of the radiation** as well as **the direction**.
- Does not need a material medium (**Fastest in vacuum**). In any medium the speed depends on the **refractive index of the medium**. Radiation is a volumetric phenomena.
- Black body**: perfect absorber $\rightarrow$ perfect emitter (at all wavelengths (**0.1 to 100 μm** (thermal radiation)), **diffuse emitter** The thermal radiation is due to the change in electronic configuration of the material due to change in its temperature.
- When the temperature of the body changes the wavelength of the emitted radiation changes as a result the color also changes



Black body inside a cavity

Thermal radiation=0.1 to 100 μm



Stefan Boltzmann Law: $\dot{Q}_{rad,Black} = \sigma AT^4$

$$\dot{Q}_{rad,A} = \epsilon_{A,T_A} \sigma A_A T_A^4 \quad \dot{Q}_{rad,B} = \epsilon_{B,T_B} \sigma A_B T_B^4$$

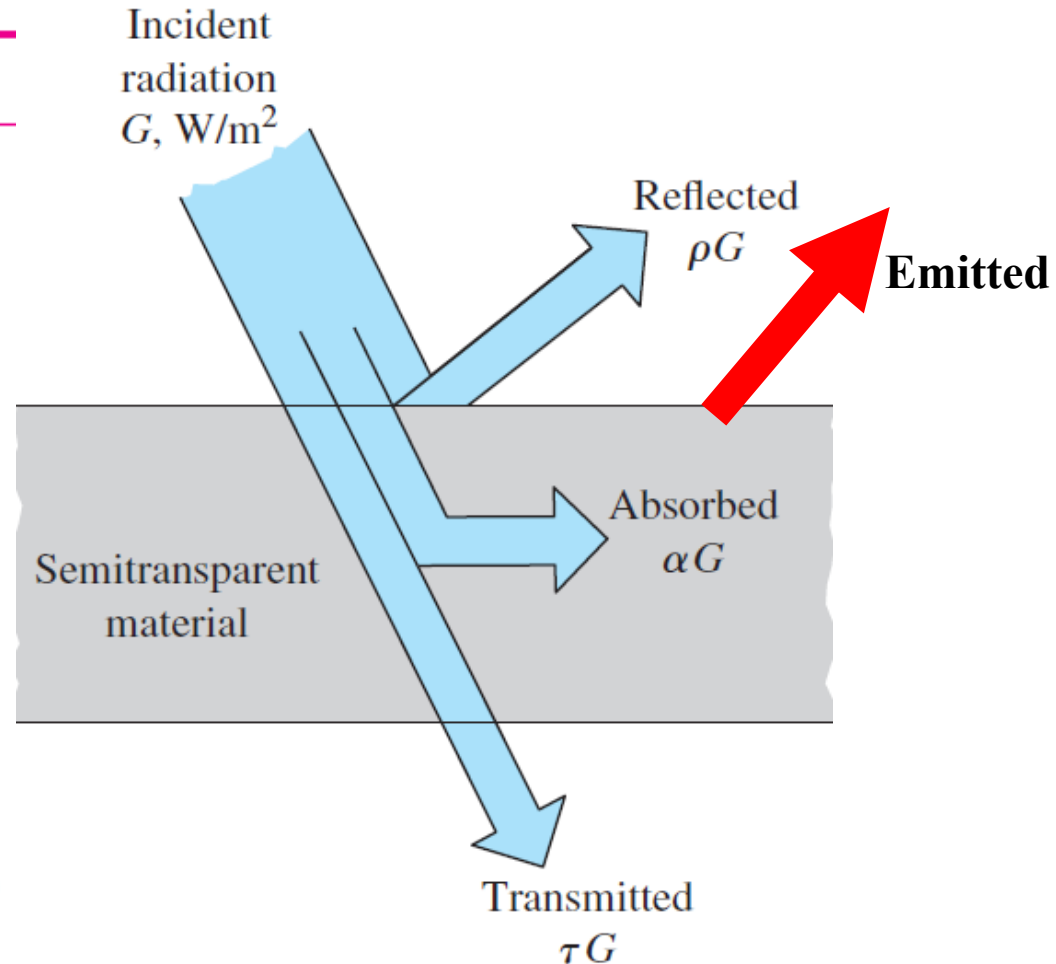
$$\dot{Q}_{A-B} = F_\epsilon \sigma A F_G (T_A^4 - T_B^4)$$

F_ϵ = Emissivity function; F_G = Geometric View Factor

Emissivity, ϵ is a measure of how closely a surface approximate a black body for which $\epsilon = 1$

Emissivity of some materials at 300 K

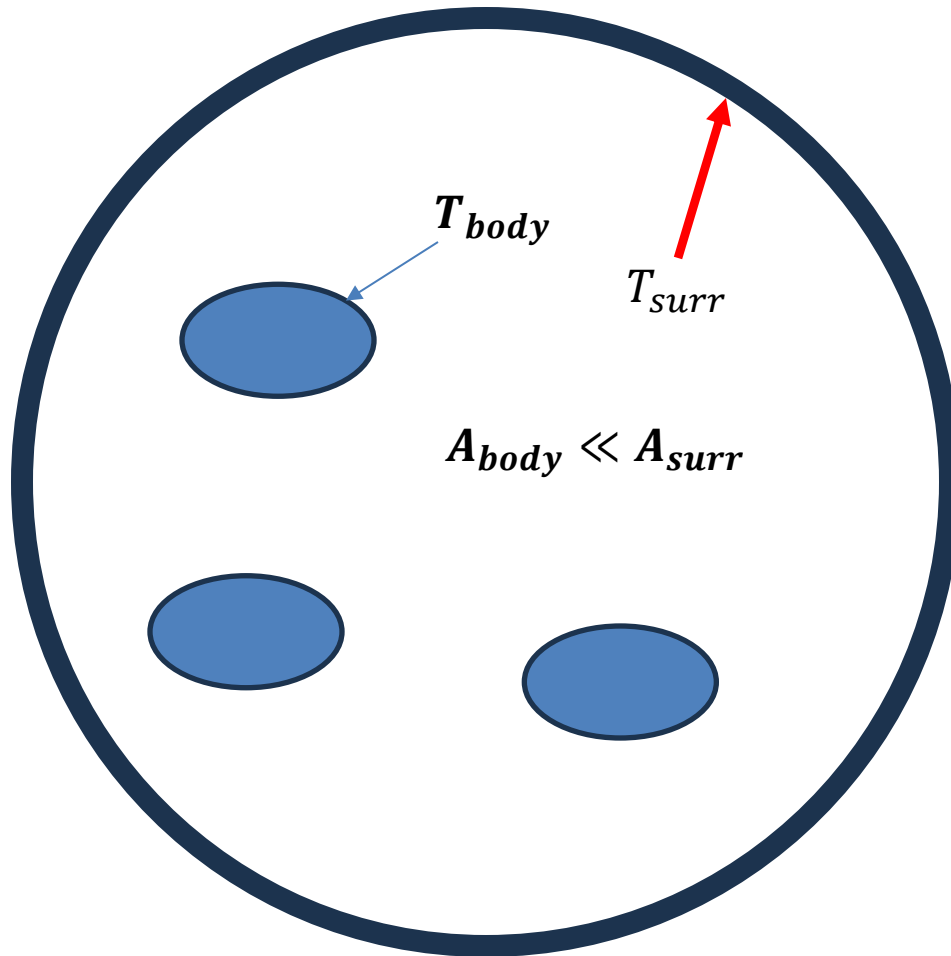
Material	Emissivity
Aluminium foil	0.07
Anodized aluminum	0.82
Polished copper	0.03
Polished gold	0.03
Polished silver	0.02
Polished stainless steel	0.17
Black paint	0.98
White paint	0.90
White paper	0.92–0.97
Asphalt pavement	0.85–0.93
Red brick	0.93–0.96
Human skin	0.95
Wood	0.82–0.92
Soil	0.93–0.96
Water	0.96
Vegetation	0.92–0.96



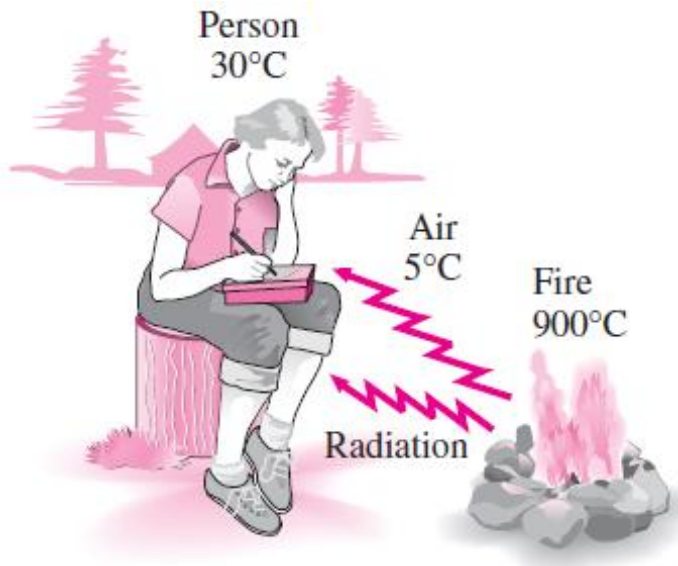
$$\rho + \alpha + \tau = 1$$

As reflectivity increases the absorptivity decreases, hence the emissivity decreases

Radiation Isotropy



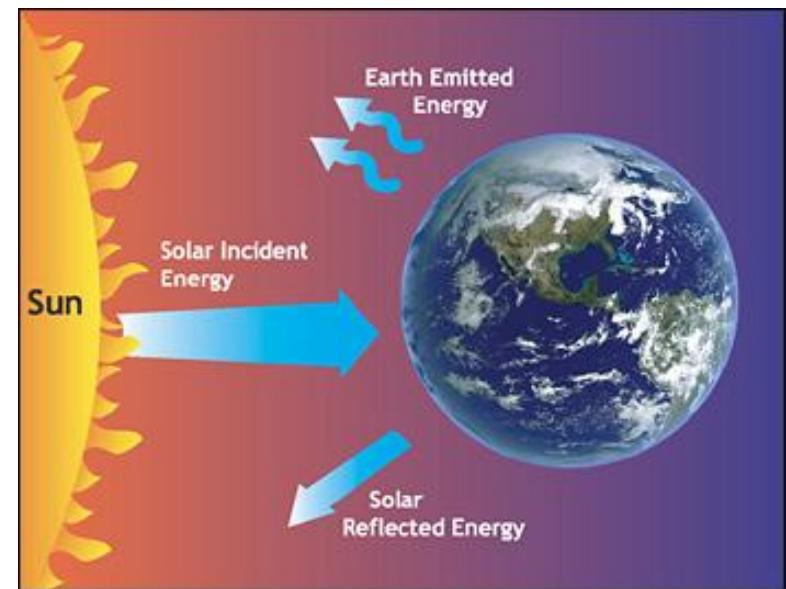
$\varepsilon = \alpha$ at thermal equilibrium



Not affected by intermediate temperature

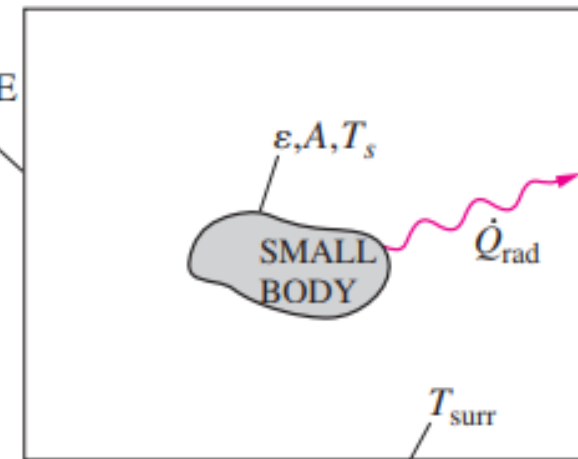
Unlike Conduction and Convection Heat transfer by radiation can occur between two bodies, even when they are separated medium colder than both of them

When a relatively small surface of emissivity ε and surface area A at absolute temperature T_s is completely surrounded by a much larger surface at absolute temperature T_{surr} , and the medium is not participating then net radiation exchange between these two surfaces is given by:

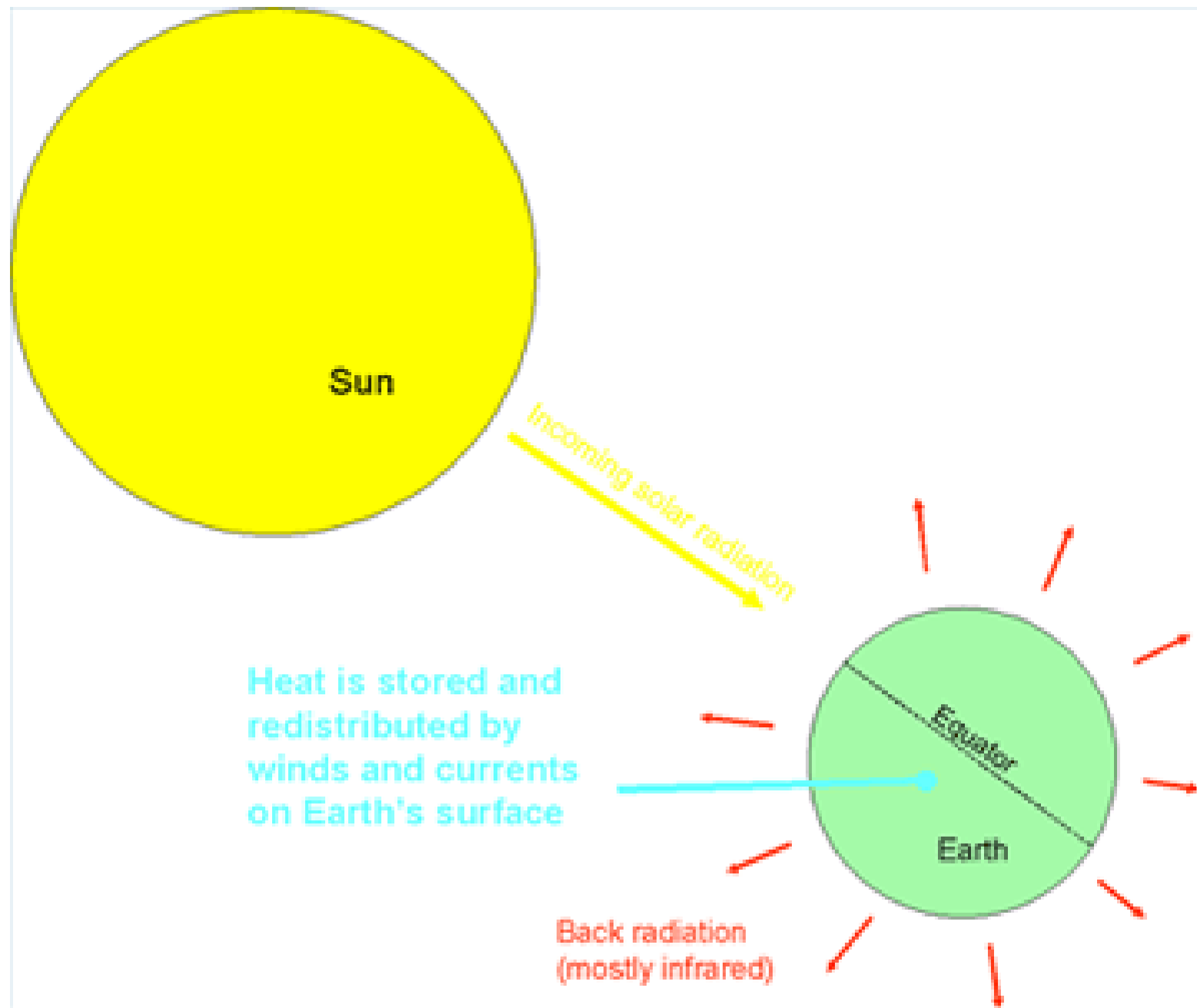


Unlike conduction and convection radiation can be possible rather better in vacuum

LARGE
ENCLOSURE



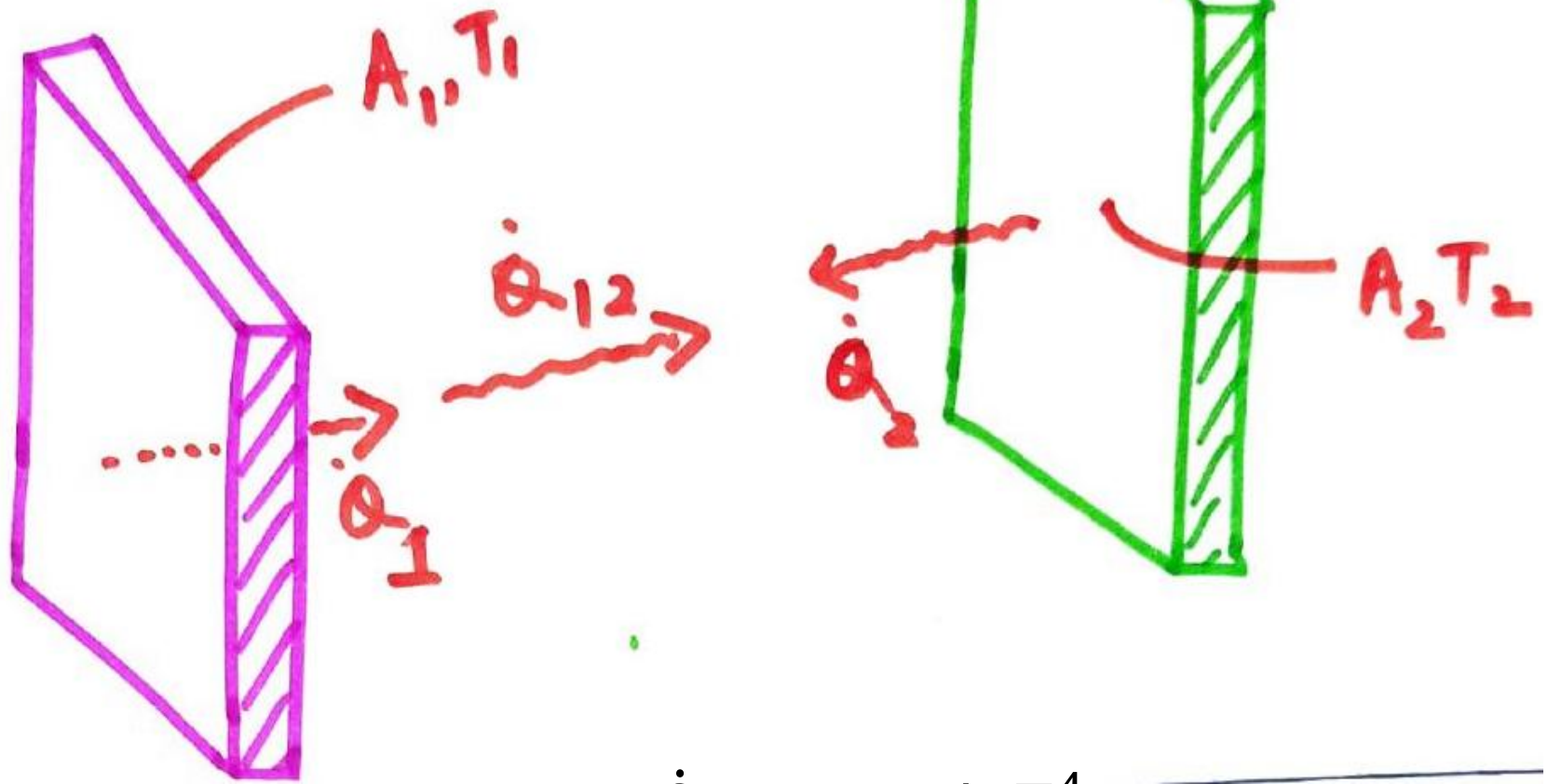
$$\dot{Q}_{rad} = \varepsilon \sigma A (T_s^4 - T_{surr}^4)$$



How to Make Things to Look Beautiful



The difference between the rates of radiation emitted by the surface and the radiation absorbed is the net radiation heat transfer.

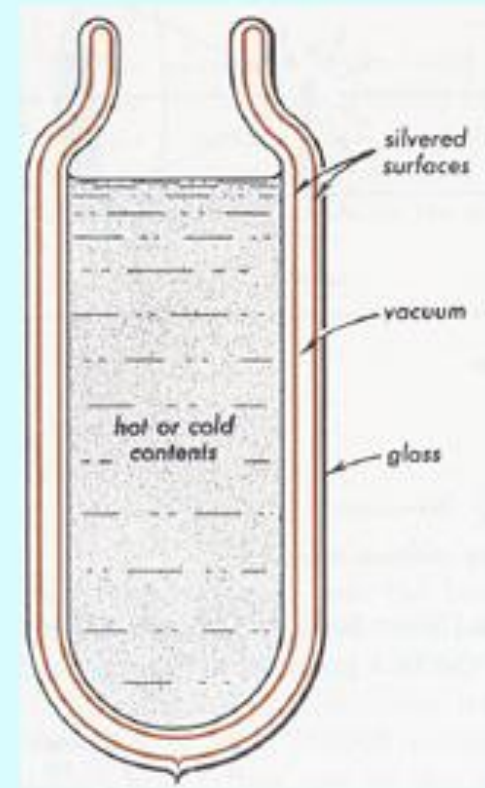


$$\dot{Q}_1 = \epsilon_1 \sigma A_1 T_1^4 \quad \dot{Q}_2 = \epsilon_2 \sigma A_2 T_2^4$$

$$\dot{Q}_{1-2} = F_{\epsilon} \sigma A F_G (T_1^4 - T_2^4)$$

Heat Transfer

Thermos Bottle



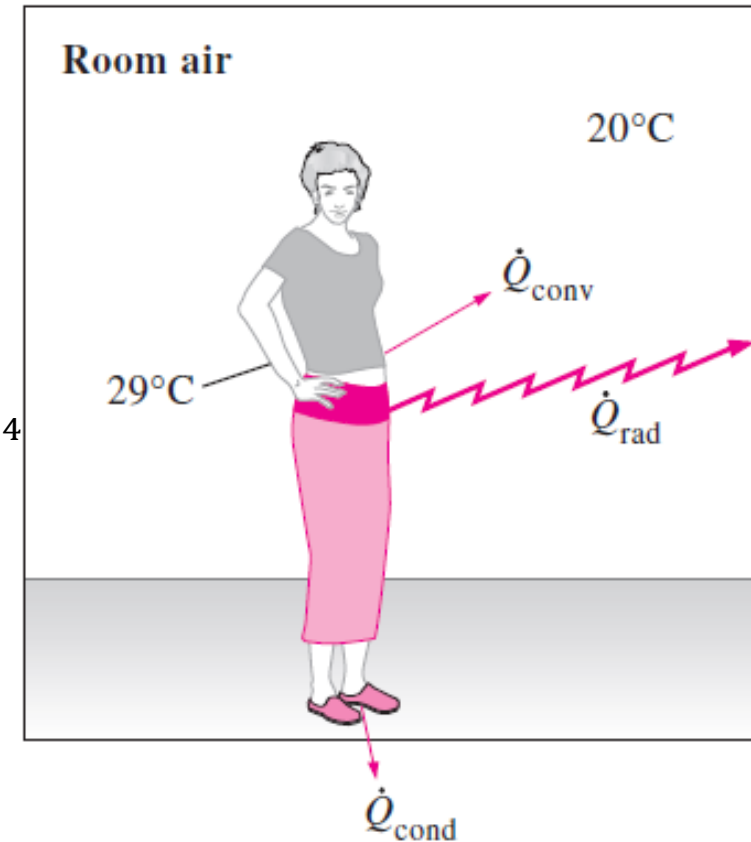
Silvered surface reflects radiated heat.
Vacuum prevents convection and conduction.
Insulated stopper reduces conduction.

Consider a person standing in a breezy room at 20°C. Determine the total rate of heat transfer from this person if the exposed surface area and the average outer surface temperature of the person are 1.6 m² and 29°C, respectively, and the convection heat transfer coefficient is 6 W/m² · °C

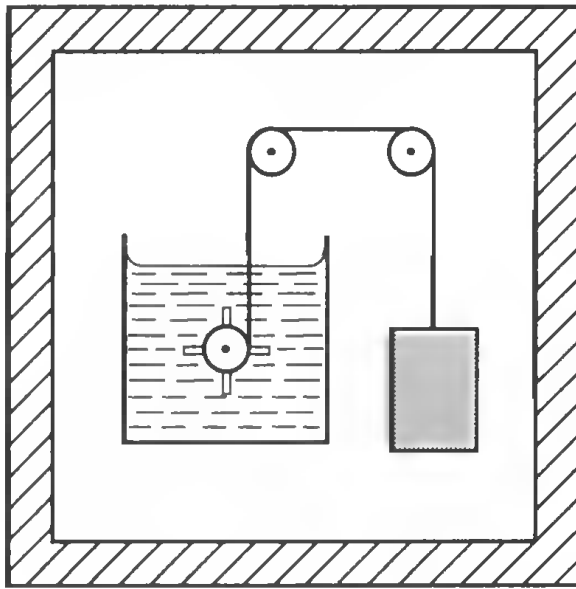
$$\begin{aligned}\dot{Q}_{con} &= hA(T_s - T_f) \\ &= \left(6 \times \frac{W}{m^2 \cdot ^\circ C}\right)(1.6 m^2)(29 - 20)^\circ C = 86.4 W\end{aligned}$$

$$\begin{aligned}\dot{Q}_{rad} &= \varepsilon\sigma A(T_s^4 - T_{surr}^4) \\ &= (0.95)(5.67 \times 10^{-8} W/m^2 K^4)(1.6 m^2)[(29 + 273)^4 \\ &\quad - (20 + 273)^4]K^4 = 81.7 W\end{aligned}$$

$$\dot{Q}_{total} = \dot{Q}_{con} + \dot{Q}_{rad} = 86.4 + 81.7 = 168.1 W$$

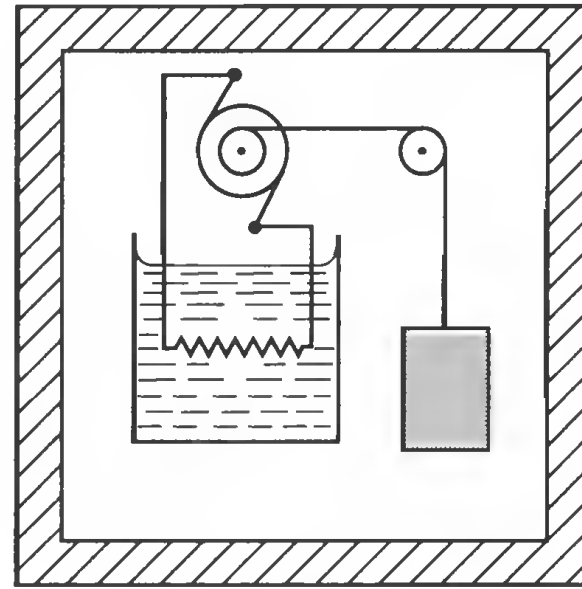


Distinction between Heat and Work

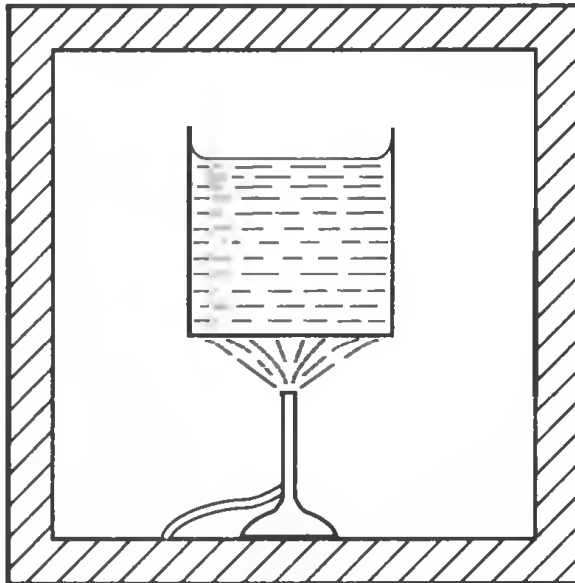


(a)

WORK

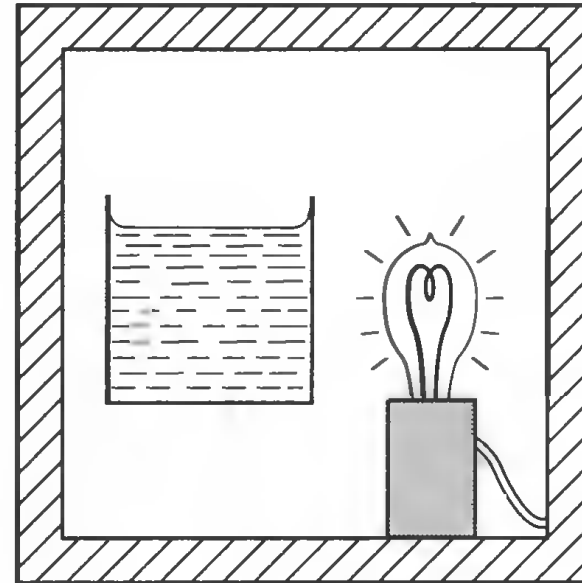


(b)



(c)

HEAT



(d)

Steady Flow

The flow through an open system is steady flow (and the system often called steady-flow system) if all the properties at each point within the system remain constant with respect to time. Property values varies from one point to another within a steady flow system but at each point they are constant with respect to time. From this definition it follows that if the flow is steady, the following must be true.

- 1) The property of the fluids crossing the boundary remains constant at each point on the boundary.
- 2) The flow rate at each section where matter crosses the boundary is constant (the flow rate can not change as long as all properties, including velocity at each point remain constant).
- 3) The rate of mass flow into the system equals the rate of mass flow out, Consequently the amount of mass within the system is constant.
- 4) The volume of the system remains constant. The system boundary must be rigid.
- 5) All interactions with the surroundings occurs at a constant rate

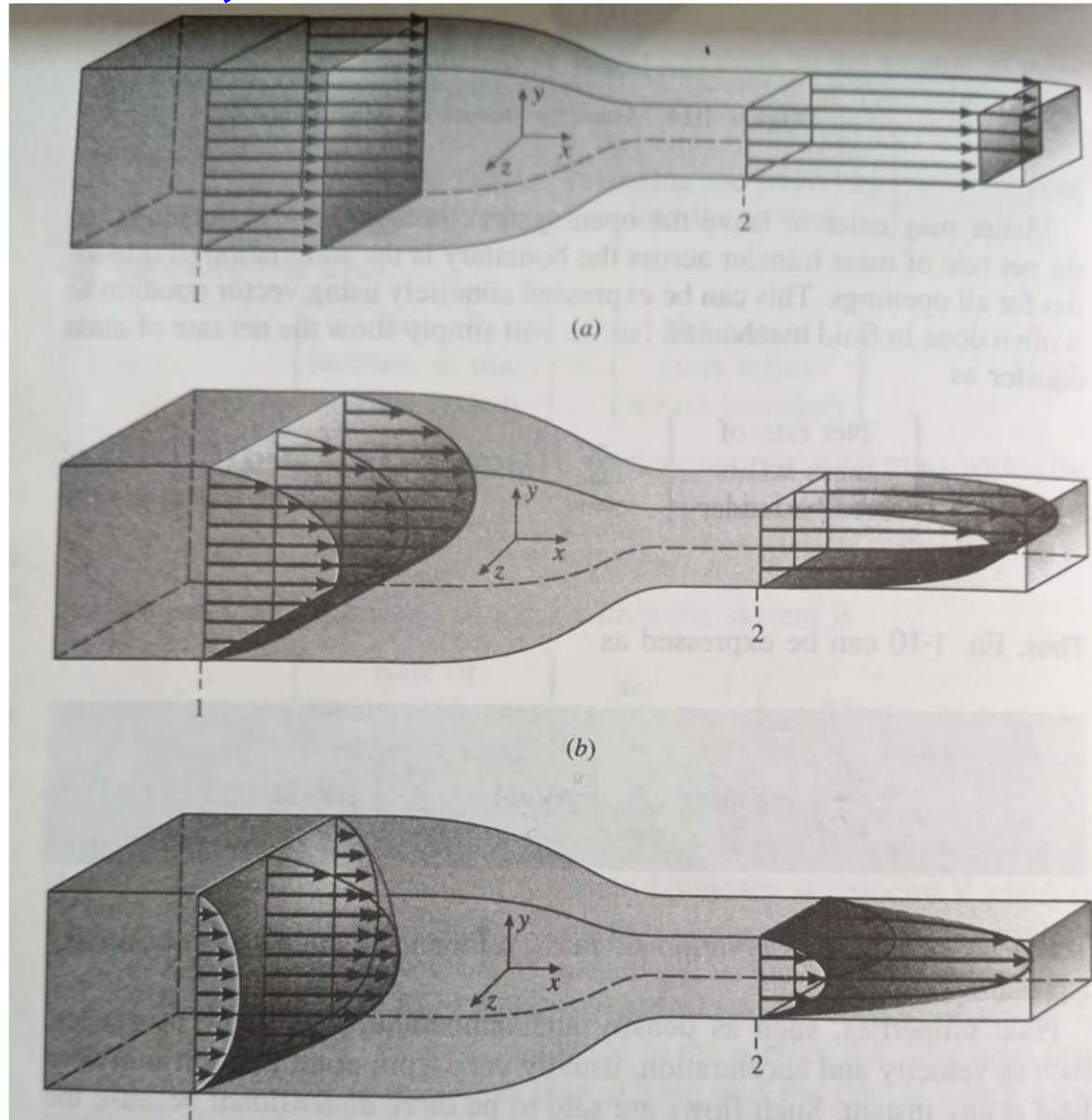
These conditions are not only necessary, they are sufficient to insure steady flow. All of these conditions are stated in terms of observations that can be made at the boundary of the system, so no knowledge of events inside a system is needed to determine if steady flow prevails. If any of these conditions is not met, the flow is unsteady or transient.

The continuity for steady flow

$$\sum_{inlets} \int \rho V dA = \sum_{outlets} \int \rho V dA$$

For one-dimensional flow $m = \rho_1 V_1 A_1 = \rho_2 V_2 A_2$

1-D, 2-D and 3-D flows

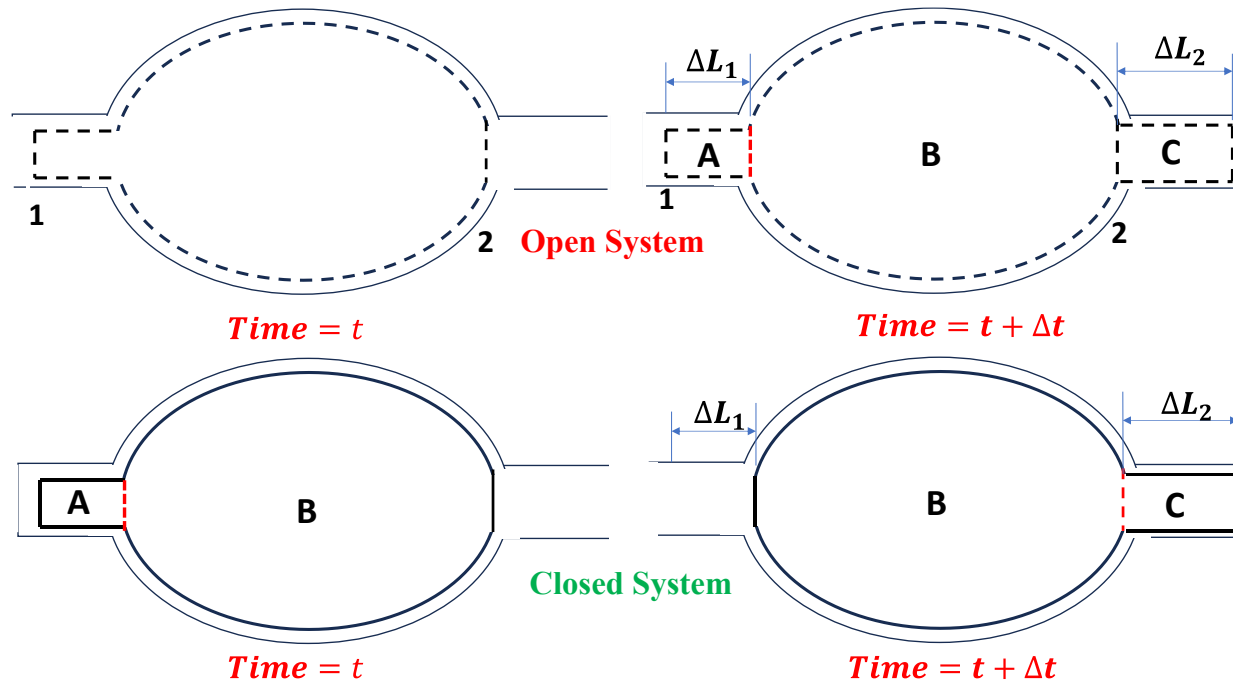


For one dimensional flow the flow properties varies only in the flow direction. $u = f(y)$, while u is the x component of the velocity is not a 1 – D flow

Work of Quasiequilibrium Process of a Simple Compressible Substance in Steady Flow

$$\text{Newton's 2}^{\text{nd}} \text{ Law : } \sum F = \frac{d}{dt}(m\mathbf{V}) \dots \dots (1)$$

Net Force on an open system/control volume in terms of the change in momentum of the fluid flowing through it:



Steady flow open system with closed system passing through it. The broken line is the open system boundary. At time t the broken line is also the boundary of the closed system. At time $t + \Delta t$ the closed system contains volumes B and C. During the time Δt , an amount of fluid of volume A flows into the open system and fluid of volume C leaves. The open system always composed of volume A and B

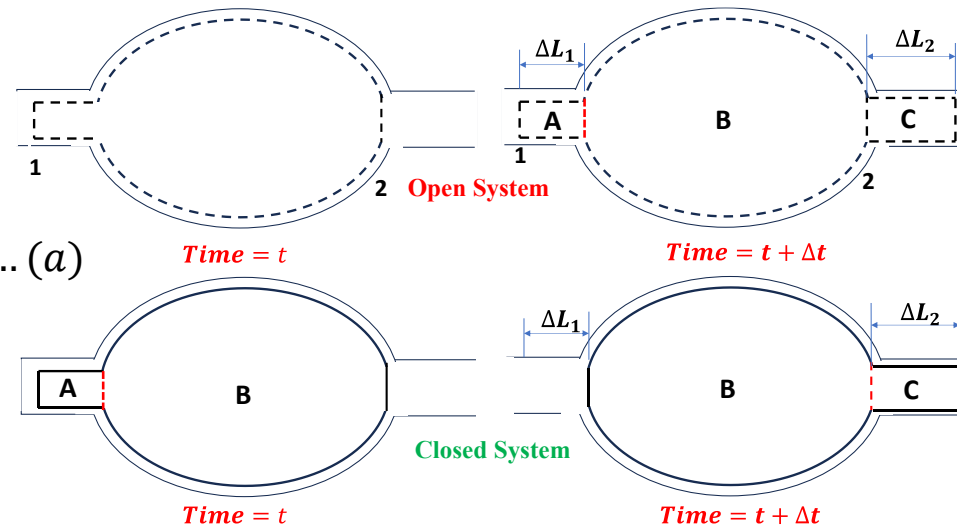
Work of Quasiequilibrium Process of a Simple Compressible Substance in Steady Flow

$$\sum F = \frac{d}{dt}(m\mathbf{V}) \dots \dots (1)$$

$$\sum F = \frac{d}{dt} \int_{\text{closed, system}} V dm$$

From the definition of derivative:

$$\sum F = \lim_{\Delta t \rightarrow 0} \frac{\left(\int_{\text{cl sys}} V dm \right)_{t+\Delta t} - \left(\int_{\text{cl sys}} V dm \right)_t}{\Delta t} \dots \dots (a)$$



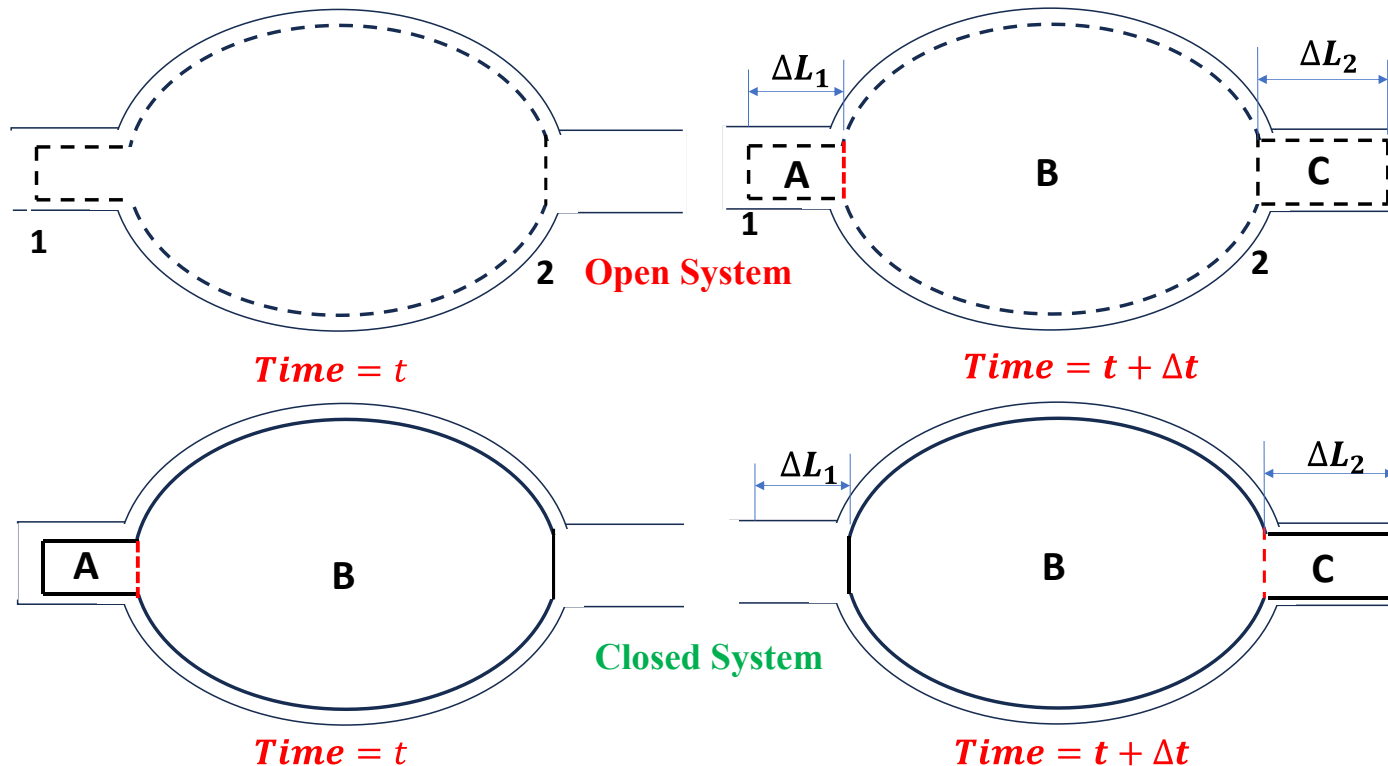
$$\left(\int_{\text{cl sys}} V dm \right)_{t+\Delta t} = \left(\int_B V dm \right)_{t+\Delta t} + \left(\int_C V dm \right)_{t+\Delta t}$$

$$= \left(\int_{\text{open sys}} V dm \right)_{t+\Delta t} - \left(\int_A V dm \right)_{t+\Delta t} + \left(\int_C V dm \right)_{t+\Delta t} \dots \dots (b)$$

$$\left(\int_{cl\ sys} V dm \right)_t = \left(\int_{open\ sys} V dm \right)_t \dots\dots(c)$$

Substituting equations b and c into a yields:

$$\sum F = \lim_{\Delta t \rightarrow 0} \frac{\left(\int_{open\ sys} V dm \right)_{t+\Delta t} - \left(\int_A V dm \right)_{t+\Delta t} + \left(\int_C V dm \right)_{t+\Delta t} - \left(\int_{open\ sys} V dm \right)_t}{\Delta t} \dots (d)$$



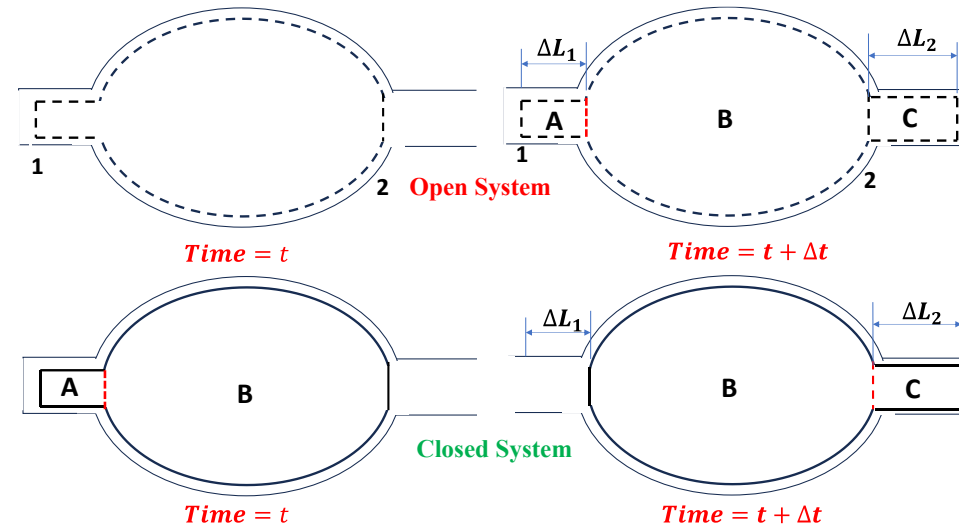
For steady flow, all properties at each point in an open system remain constant with respect to time, so the 1st and last terms in equation d are equal in magnitude

$$\sum F = \lim_{\Delta t \rightarrow 0} \frac{\left(\int_C V dm \right)_{t+\Delta t} - \left(\int_A V dm \right)_{t+\Delta t}}{\Delta t} \dots (e)$$

Since the limit of a sum equals the sum of the limits

$$\sum F = \lim_{\Delta t \rightarrow 0} \frac{\left(\int_C V dm \right)_{t+\Delta t}}{\Delta t} - \lim_{\Delta t \rightarrow 0} \frac{\left(\int_A V dm \right)_{t+\Delta t}}{\Delta t} \dots (f)$$

Lets now restrict the flow across section 2 to One-dimensional flow with **V perpendicular to A_2** .
The 1st term on the righthand side of equation f can be simplified as follows



$$\text{Term 1} = \lim_{\Delta t \rightarrow 0} \frac{\left(\int_C V dm \right)_{t+\Delta t}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\left(\int_{A_2} V \rho \Delta L_2 dA \right)_{t+\Delta t}}{\Delta t}$$

$$Term\ 1 = \lim_{\Delta t \rightarrow 0} \frac{\left(\int_C V dm \right)_{t+\Delta t}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\left(\int_{A_2} V \rho \Delta L_2 dA \right)_{t+\Delta t}}{\Delta t}$$

Where $dm = \rho dVol = \rho \Delta L_2 dA$ integration is over A_2 , since all the fluid in volume C at time $t + \Delta t$ passes through area A_2 during the time interval. Further remembering that the flow is steady and one dimensional and then passing to the limit, we find that

$$Term\ 1 = \lim_{\Delta t \rightarrow 0} \frac{\left(\int_{A_2} V \rho \Delta L_2 dA \right)_{t+\Delta t}}{\Delta t} = V_2 \dot{m}_2$$

In a similar manner

$$Term\ 2 = \lim_{\Delta t \rightarrow 0} \frac{\left(\int_A V dm \right)_{t+\Delta t}}{\Delta t} \lim_{\Delta t \rightarrow 0} \frac{\left(\int_{A_1} V \rho \Delta L_1 dA \right)_{t+\Delta t}}{\Delta t} = V_1 \dot{m}_1$$

For steady flow with single inlet and exit, $\dot{m}_1 = \dot{m}_2 = \dot{m}$

$$\sum \mathbf{F} = \dot{m}(\mathbf{V}_2 - \mathbf{V}_1)$$

It is a vector equation, although for our present purpose a special case of colinear V_1 and V_2 is adequate.

Work of Quasiequilibrium Process of a Simple Compressible Substance in Steady Flow

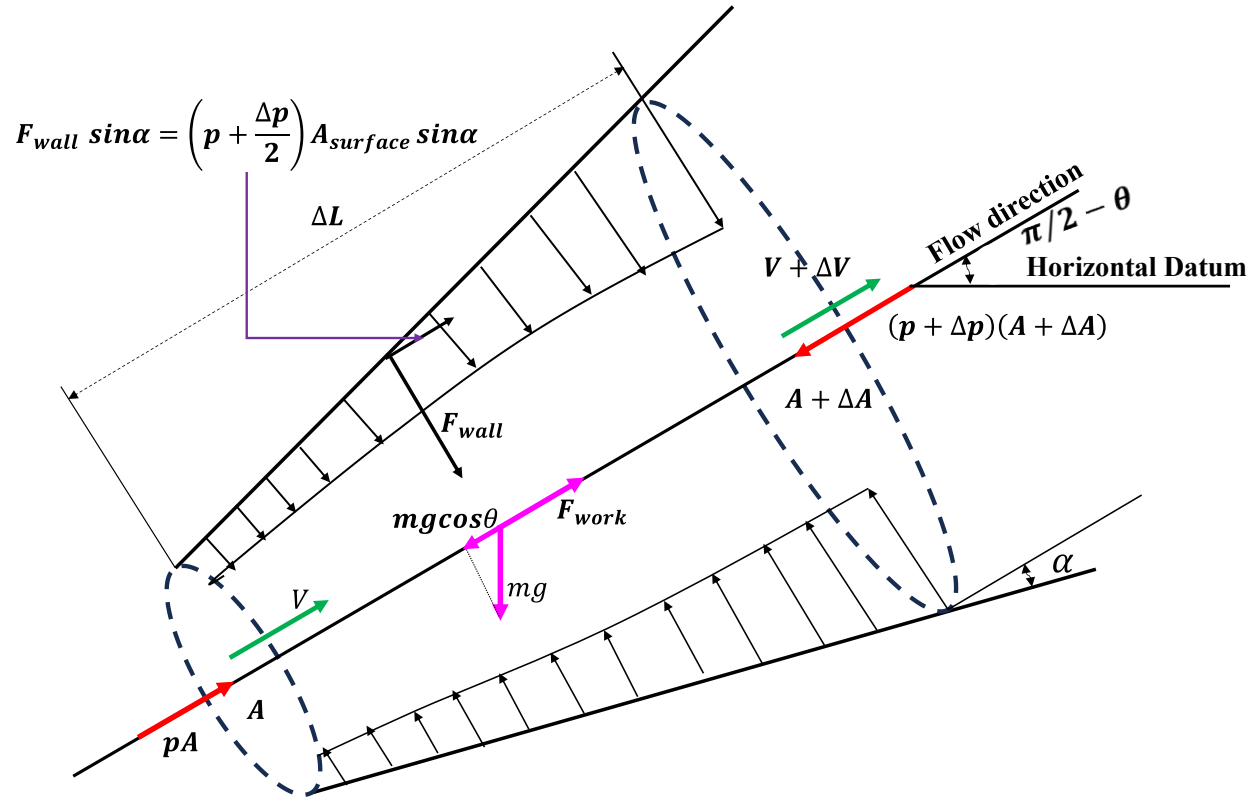
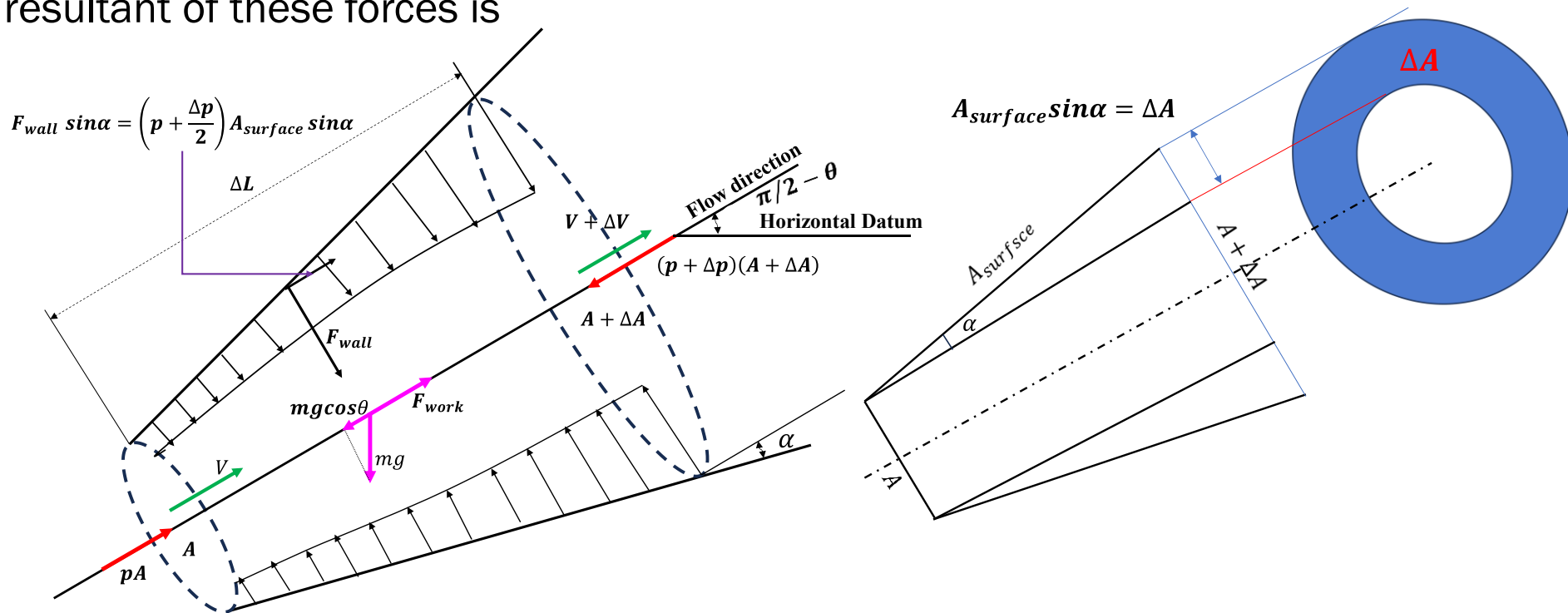


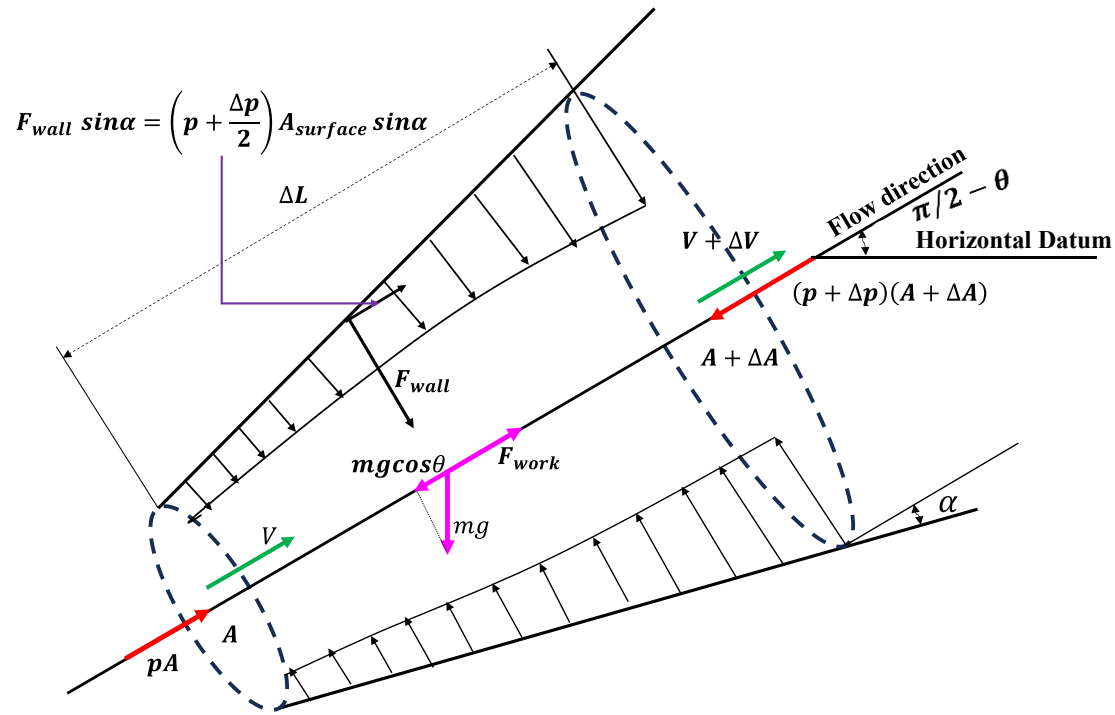
Figure shows a small steady flow system that is a part of large system. The length ΔL is small enough so that the variation of p , A , and V along ΔL are very nearly linear. This does not mean that p , A , and V vary linearly with L , but only that ΔL is so small that a linear relation is a good approximation for the short distance ΔL . This approximation becomes better as ΔL becomes smaller, and later in the derivation an exact expression will be obtained by letting ΔL approach the differential length dL .

1. The force of the adjacent fluid on the inlet area, pA
2. The force of the adjacent fluid on the inlet area, $(p + \Delta p)(A + \Delta A)$
3. The component of the weight of the element, $mg\cos\theta$
4. The component of the normal wall force F_{wall} in the direction of flow

$$F_{wall}\sin\alpha = \left(p + \frac{\Delta p}{2}\right) A_{surface}\sin\alpha = \left(p + \frac{\Delta p}{2}\right) \Delta A$$

The force F_{work} by which work is done on the fluid. This force is applied to the fluid by means of some impeller that is not shown in the figure. The sum of the resultant of these forces is





$$\sum F = pA - (p + \Delta p)(A + \Delta A) - \rho \left(A + \frac{\Delta A}{2}\right) \Delta L g \cos \theta + \left(p + \frac{\Delta p}{2}\right) \Delta A + F_{work}$$

$$\sum F = -A\Delta p - \frac{\Delta p \Delta A}{2} - \rho A \Delta L g \cos \theta + \rho g \frac{\Delta A \Delta L}{2} \cos \theta + F_{work}$$

$$\sum F = -A\Delta p - \frac{\Delta p \Delta A}{2} - \rho A \Delta L g \cos \theta + \rho g \frac{\Delta A \Delta L}{2} \cos \theta + F_{work} = \dot{m}(V + \Delta V - V)$$

$$F_{work} = \dot{m} \Delta V + A \Delta p + \rho A \Delta L g \cos \theta$$

F_{work} is in the direction of flow. This force acting on the fluid as it moves the distance ΔL does work on the fluid. So, the amount of work is

$$W_{in} = F_{work}\Delta L = \rho AV\Delta V\Delta L + A\Delta p\Delta L + \rho A\Delta L g \cos\theta \Delta L$$

Divide by the mass of the fluid in the open system, $\rho A\Delta L$ to obtain the work input per unit of the mass

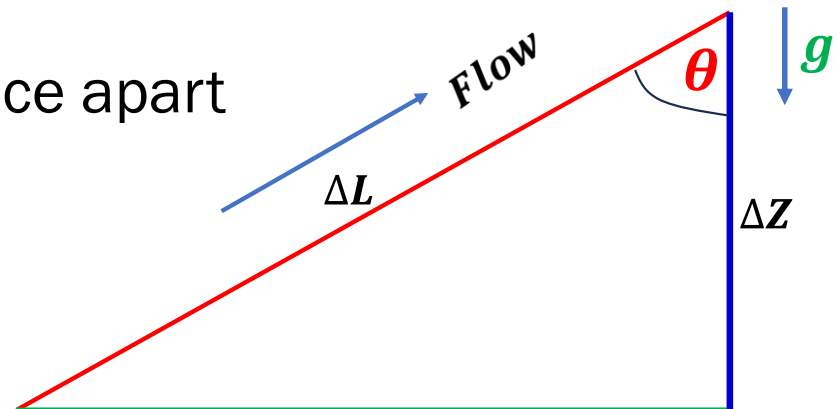
$$w_{in} = V\Delta V + \frac{\Delta p}{\rho} + g \cos\theta \Delta L = v\Delta p + V\Delta V + g \cos\theta \Delta L$$

ΔL is made to approach dL then the other differences also approach differentials, and the work per unit mass done on the fluid in the distance dL is

$$\delta w_{in} = v dp + V dV + g \cos\theta dL$$

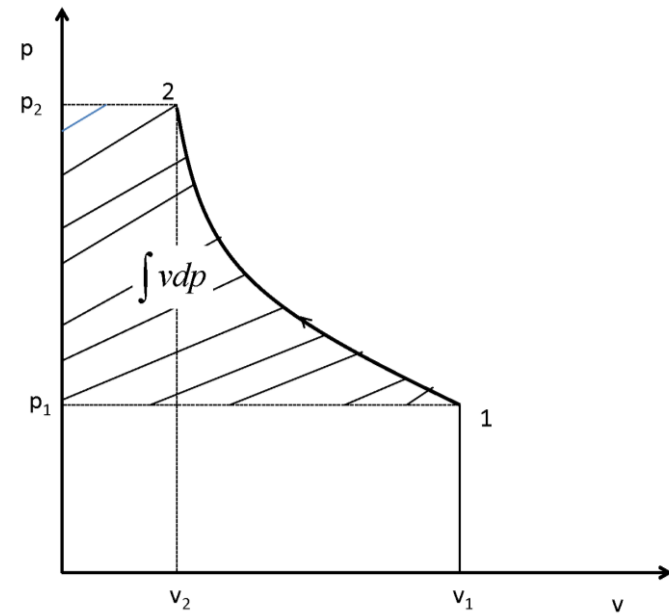
Flow between sections a finite distance apart

$$w = - \int v dp - \Delta \left(\frac{V^2}{2} \right) - g \Delta Z$$



Work of a **quasiequilibrium** and **frictionless** process of a **simple compressible substance** in **one dimensional steady flow** (Actually the derivation requires that the **flow is one dimensional at the inlet and exit, not throughout the open system**) is given by:

$$W_{flow,rev} = - \int v dp - \Delta \frac{V^2}{2} - g \Delta z$$

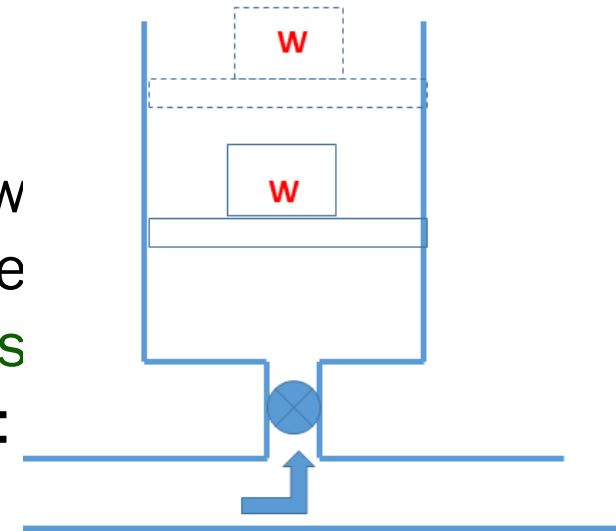


For reversible steady state process involving **negligible change in KE & PE** the **shaft work per unit mass** is given by :

$$W_{flow,rev} = - \int v dp$$

Imp. Note: Suppose in an **open system** (flow process) there is **moving boundary work**. If the process is **quasistatic** as well as **frictionless** then the moving boundary work will be given by:

$$W_{movingboundary,rev} = \int p dV$$



Note that: $\int v dp$ is not the same as expansion work $\int p dv$

We can find a relation between reversible shaft work and reversible expansion work from elementary calculus:

$$\int p dv = \int d(pv) - \int v dp$$

 Reversible expansion work  Flow work  Reversible shaft work
